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Pediatric advanced life support (PALS)

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INTRODUCTION

This topic will discuss the advanced components of recognition and treatment of respiratory failure, shock, cardiopulmonary failure, and cardiac arrhythmias in children.

Basic life support and post-cardiac arrest care in children are discussed separately:

- (See "Pediatric basic life support (BLS) for health care providers".)
- (See "Initial post-cardiac arrest care in children".)

THE PALS PROGRAM

The American Heart Association (AHA) PALS program provides a structured approach to the assessment and treatment of the critically ill pediatric patient [1,2]. The guidelines are based upon the International Liaison Committee on Resuscitation (ILCOR) consensus on science, which performs a continuous evidence evaluation process and publishes periodic focused updates to the AHA PALS guidelines. The new Pediatric Advanced Life Support algorithms are available here.

The PALS content includes:

- Overview of assessment
- Recognition and management of respiratory distress and failure
- Recognition and management of shock
- Recognition and management of cardiac arrhythmias
- Recognition and management of cardiac arrest
- Postresuscitation management of patients with pulmonary and cardiac arrest
- Review of pharmacology

PREVENTION OF CARDIAC ARREST

The clinician should primarily focus on prevention of cardiopulmonary failure through early recognition and management of respiratory distress, respiratory failure, and shock that can lead to cardiac arrest due to progressive hypoxia, acidosis, and ischemia [2]. Pediatric early warning scores and rapid response teams (RRTs) provide institutional support for early recognition of deterioration in hospitalized children.

Pediatric early warning scores — For hospitalized children, we suggest use of validated pediatric early warning scores (PEWS) to identify those patients whose condition may be worsening [3]. PEWS are frequently used by hospitals that care for children in conjunction with pediatric RRTs to monitor the behavioral, cardiovascular, respiratory status and degree of medical support of their pediatric patients during hospitalization. Increasing PEWS suggest that a child's clinical condition is worsening. PEWS scoring systems typically recommend interventions based on thresholds that suggest or require specific actions. Several validated PEWS are in use [3,4].

In an international randomized trial of 21 pediatric hospitals that compared use of the Bedside Paediatric Early Warning System (BedsidePEWS) with no severity of illness score over 550,000 patient-days, hospitals that implemented the BedsidePEWS had

fewer significant clinical deterioration events such as emergency endotracheal intubation, fluid resuscitation, or initiation of CPR (adjusted relative risk [aRR] 0.77, 95% CI 0.61-0.97) but no significant difference in all-cause mortality [4,5]. In the 2022 Consensus on Science systematic review and meta-analysis of PEWS (12 studies), in addition to the trial mentioned above, pooled results from cohort studies found similar results [3]. However, in all studies, the rates of mortality were low, and although not statistically significant, mortality may be increased in hospitals that do not use PEWS.

Rapid response teams — For hospitals that care for children, we suggest use of a pediatric rapid response team (RRT). This recommendation places a high value on preventing clinical deterioration in children despite the resources needed to implement these systems. A RRT, also known as a medical emergency team (MET), consists of personnel from medical, nursing and/or respiratory therapy who have critical care training and are available 24 hours per day, seven days a week for evaluation and treatment of patients who are located in non-critical care settings (eg, medical or surgical inpatient wards) and show signs of clinical deterioration. Implementation of an RRT has been promoted as a major strategy for improving patient safety in hospitals [6]. Infants and children with high-risk conditions who are managed on general inpatient units may benefit from RRTs that can provide prompt assessment and management if clinical deterioration occurs although results from large studies are not consistent [1,7-10]:

- A meta-analysis of five pediatric prospective observational studies with a total of 347,618 patient admissions found that implementation of an RRT was associated with a significant reduction in deaths from cardiac arrest when compared with historical control periods (0.05 versus 0.17 percent, relative risk [RR] 0.6, 95% CI 0.5-0.8) [7]. However, decreased mortality after implementation of an RRT was not found in all studies.
- A cohort study of 29,294 patient admissions (7257 admissions after institution of an RRT) that was included in the meta-analysis compared hospital-wide mortality rates and rates of respiratory and cardiopulmonary arrests outside of the intensive care unit before and after implementation of an RRT in a 264-bed freestanding children's hospital [8]. Major findings included:
  - The mean monthly mortality rate decreased from 1 to 0.8 deaths per 100 discharges (18 percent decrease, 95% CI 5-30 percent).
  - The mean monthly code rate (respiratory or cardiopulmonary arrest) decreased from 2.5 to 0.7 codes per 1000 patient admissions (RR 0.3, 95% CI 0.1-0.7). A possible explanation for this finding is that early activation of the RRT in a critically ill patient might have prevented codes.
  - Over 18 months, the RRT was activated 143 times, most commonly for respiratory distress, hypotension, hypoxemia, altered mental status, and tachycardia. The most common actions by the RRT were respiratory support, fluid resuscitation, airway management, and transfer to the intensive care unit.
- A multicenter, prospective observational study of the implementation of a clinician-led pediatric RRT in four pediatric academic centers found that initiation of an RRT was associated with a significant reduction in pediatric intensive care unit mortality rate after readmission from a medical or surgical unit (0.3 to 0.1 deaths per 1000 hospital admissions) but no significant decline in the rate of cardiopulmonary arrests [9].

However, these observations do not prove that the RRT was responsible for the improvement in outcomes. Support for this concern comes from an observational study in a children's hospital that did **not** implement an RRT but also found a significant reduction in mortality over the same time period in which other pediatric centers reported decreased mortality in association with RRT implementation [10]. In a separate multicenter study that evaluated predicted versus actual hospital mortality rates in 38 free-standing children's hospitals before and after implementation of an RRT and that controlled for declining mortality over time, RRT implementation was not associated with a reduction in hospital deaths beyond what was predicted by preimplementation trends [11].

Thus, the benefit of an RRT is not consistent across all settings, and it is possible that explanations other than the RRT may be responsible for at least part of the benefit. The quality and generalizability of the evidence describing the effectiveness of implementing RRTs is also limited by features such as before and after observation design, selection of primary and secondary outcome measures, and varied indications for RRT activation. In addition, because the mortality following pediatric intensive care unit (PICU) admission is typically low, its utility as an outcome measure may be limited. Finally, the systems being studied are complex, making it difficult to identify confounding factors such as changes in secular trends or indirect benefits derived from the RRT implementation. Thus, institutions may choose to implement and maintain RRTs based upon their own safety priorities and resources.

ASSESSMENT

The assessment of respiratory distress and circulatory compromise in children, including the common findings, is covered in greater detail separately. (See "Initial assessment and stabilization of children with respiratory or circulatory compromise".)

PALS uses an assessment model that facilitates rapid evaluation and intervention for life-threatening conditions. In infants and children, most cardiac arrests result from progressive respiratory failure and/or shock, and one of the aims of this rapid assessment model is to prevent progression to cardiac arrest.

The evaluation includes:

- Initial impression (brief visual and auditory observation of child's overall appearance, work of breathing, circulation) (see "Initial assessment and stabilization of children with respiratory or circulatory compromise", section on 'Pediatric assessment triangle')
- Primary assessment – The clinician should in rapid sequence assess:
  - Airway (patent, patent with maneuvers/adjuncts, partially or completely obstructed)
  - Breathing (respiratory rate, effort, tidal volume, lung sounds, pulse oximetry)
  - Circulation (skin color and temperature, heart rate and rhythm, blood pressure, peripheral and central pulses, capillary refill time)
  - Disability
    - AVPU pediatric response scale: Alert, Voice, Pain, Unresponsive
    - Pupillary response to light
    - Presence of hypoglycemia (rapid bedside glucose or response to empiric administration of dextrose)
    - Glasgow Coma Scale: Eye Opening, Verbal Response, Motor Response (table 1) (for trauma patients)
  - Exposure (fever or hypothermia, skin findings, evidence of trauma)
- Secondary assessment – This portion of the evaluation includes a thorough head to toe physical examination, as well as a focused medical history that consists of the "SAMPLE" history:
  - **S:** Signs and symptoms
  - **A:** Allergies
  - **M:** Medications
  - **P:** Past medical history
  - **L:** Last meal
  - **E:** Events leading to current illness
- Tertiary assessment – Injury and infection are common causes of life-threatening illness in children. Thus, ancillary studies are frequently directed towards identifying the extent of trauma or an infectious focus. (See "Trauma in children: Initial management of the unstable patient (primary survey)", section on 'Adjuncts' and "Initial evaluation of shock in children", section on 'Evaluation' and "Approach to the ill-appearing infant (younger than 90 days of age)", section on 'Ancillary studies for infectious etiologies' and "Trauma in children: Secondary survey (evaluation)", section on 'Adjuncts'.)

Respiratory distress and failure — Recognition and treatment of respiratory conditions that respond to simple measures (eg, supplemental oxygen or inhaled bronchodilators) are major goals of PALS [12]. The clinician may also have to treat rapidly progressive conditions and intervene with advanced therapies to avoid cardiopulmonary arrest in patients with respiratory failure. Early detection and treatment improve overall outcome.

There are many causes of acute respiratory compromise in children (table 2). The clinician should strive to categorize respiratory distress or failure into one or more of the following [12] (see "Acute respiratory distress in children: Emergency evaluation and initial stabilization"):

- Upper airway obstruction (eg, croup, epiglottitis)
- Lower airway obstruction (eg, bronchiolitis, status asthmaticus)
- Lung tissue (parenchymal) disease (eg, bronchopneumonia)
- Disordered control of breathing (eg, seizure, coma, muscle weakness)

Initial management supports airway, breathing, and circulation:

- **Airway** – Key steps in basic airway management include (see "Basic airway management in children"):
  - Provide 100 percent inspired oxygen
  - Allow the child to assume a position of comfort or manually open the airway
  - Clear the airway (suction)
  - Insert an airway adjunct if consciousness is impaired (eg, nasopharyngeal airway or, if gag reflex absent, oropharyngeal airway)
  - If the patient is unresponsive without signs of life, begin chest compressions immediately
- **Breathing** – The clinician should:
  - Assist ventilation manually in patients not responding to basic airway maneuvers or with inadequate or ineffective respiratory effort
  - Monitor oxygenation by pulse oximetry
  - Monitor ventilation by end-tidal carbon dioxide ($EtCO_2$) if available
  - Administer medications as needed (eg, albuterol for status asthmaticus, inhaled racemic epinephrine for croup)

  In preparation for intubation, 100 percent oxygen should be applied via non-rebreather mask or other high concentration device. If the patient has evidence of respiratory failure, positive pressure ventilation should be initiated with a bag-valve-mask or flow-inflating device to oxygenate and improve ventilation. (See "Basic airway management in children".)

  Children who cannot maintain their airway, oxygenation, or ventilatory requirements should undergo placement of an artificial airway, usually via endotracheal intubation and, less commonly, with a supraglottic airway or alternative device. Certain populations of patients with upper airway obstruction and/or respiratory failure may respond to noninvasive ventilation (CPAP or BiPAP) if airway reflexes are preserved. A rapid overview describes the steps in performing rapid sequence intubation (table 3). (See "Noninvasive ventilation for acute and impending respiratory failure in children" and "Technique of emergency endotracheal intubation in children" and "Rapid sequence intubation (RSI) in children for emergency medicine: Approach".)
- **Circulation** – Key interventions consist of monitoring heart rate and rhythm and establishing vascular access to provide volume administration and/or medications for resuscitation. (See "Emergency and elective venous access in children".)

Shock — The goal is to recognize and categorize the type of shock in order to prioritize treatment options (algorithm 1). Early treatment of shock may prevent the progression to cardiopulmonary failure (algorithm 2). The management of shock is discussed separately. (See "Shock in children in resource-abundant settings: Initial management" and "Children with sepsis in resource-abundant settings: Rapid recognition and initial resuscitation (first hour)".)

Shock may occur with normal, increased, or decreased systolic blood pressure. Shock in children is usually related to low cardiac output, but some patients may have high cardiac output, such as with sepsis or severe anemia. (See "Initial evaluation of shock in children".)

Shock severity is usually categorized by its effect on systolic blood pressure [13]:

- **Compensated shock** – Compensated shock occurs when compensatory mechanisms (including tachycardia, increased systemic vascular resistance, increased inotropy, and increased venous tone) maintain a systolic blood pressure within a normal range (table 4 and table 5).
- **Hypotensive shock (or decompensated shock)** – Hypotensive shock occurs when compensatory mechanisms fail to maintain systolic blood pressure as indicated by a systolic blood pressure <5th percentile for age [13]:

  The PALS definition of systolic hypotension varies by age:

  - Term neonates (0 to 28 days): <60 mmHg
  - Infants (1 to 12 months): <70 mmHg
  - Children (1 to 10 years): <70 mmHg + (child's age in years x 2)
  - Children >10 years: <90 mmHg
- **Shock categorization** – There are four major categories of shock [13] (see "Initial evaluation of shock in children"):

- **Hypovolemic shock** – Hypovolemic shock is characterized by inadequate circulating blood volume. Common causes of fluid loss include diarrhea, hemorrhage (internal and external), vomiting, inadequate fluid intake, osmotic diuresis (eg, diabetic ketoacidosis), third-space losses, and burns.

- **Distributive shock** – Distributive shock describes inappropriately distributed blood volume typically associated with decreased systemic vascular resistance. Common causes include septic shock, anaphylactic shock, and neurogenic shock (eg, head injury, spinal injury).

- **Cardiogenic shock** – Cardiogenic shock refers to impairment of heart contractility. Common causes include congenital heart disease, myocarditis, cardiomyopathy, arrhythmias, sepsis, poisoning or drug toxicity, and myocardial injury (trauma).

- **Obstructive shock** – In this form of shock, hypotension arises from obstructed blood flow to the heart or great vessels. Common causes include cardiac tamponade, tension pneumothorax, ductal dependent congenital heart lesions, and massive pulmonary embolism.

  Any given patient may suffer from more than one type of shock. For example, a child in septic shock may develop hypovolemia during the prodrome phase, distributive shock during the early phase of sepsis, and cardiogenic shock later in the course.

Cardiopulmonary failure — Respiratory failure and hypotensive shock are the most common conditions preceding cardiac arrest in children [2].

- Causes of respiratory failure include:

  - Upper airway obstruction (choking, infection)
  - Lower airway obstruction (asthma, foreign body aspiration)
  - Parenchymal disease (pneumonia, acute pulmonary edema)
  - Disordered control of breathing (coma, toxic ingestion, status epilepticus)

- Causes of hypotensive shock include:

  - Hypovolemia (dehydration, hemorrhage)
  - Cardiac failure (eg, due to myocarditis or valvular disease)
  - Distributive shock (septic, neurogenic)
  - Metabolic/electrolyte disturbances
  - Acute myocardial infarction/ischemia
  - Toxicologic ingestions
  - Pulmonary embolism

- The following physical findings often precede cardiopulmonary failure:

  - **Airway** – Stridor, stertor, drooling, and/or severe retractions

  - **Breathing** – Bradypnea, irregular, ineffective respiration, gasping, and/or cyanosis

  - **Circulation** – Bradycardia, capillary refill >5 seconds, weak central pulses, no peripheral pulses, hypotension, cool extremities, and/or mottled/cyanotic skin

  - **Disability** – Diminished level of consciousness

The patient in cardiopulmonary failure will progress rapidly to cardiac arrest without aggressive intervention. Positive pressure ventilations with 100 percent inspired oxygen, chest compressions for heart rate <60 beats per minute in patients with poor perfusion, and administration of intravenous fluids and medications tailored to treat the underlying cause are indicated. (See "Basic airway management in children" and "Pediatric basic life support (BLS) for health care providers".)

Heart rate and rhythm — In children, the heart rate is classified as bradycardia, tachycardia, and pulseless arrest. Interpretation of the cardiac rhythm requires knowledge of the child's typical heart rate (table 6) and baseline rhythm as well as level of activity and clinical condition.

Bradycardia — Bradyarrhythmias are common pre-arrest rhythms in children and are often due to hypoxia. Bradycardia with symptoms of shock (eg, poor systemic perfusion, hypotension, altered consciousness) requires urgent treatment to prevent cardiac arrest (algorithm 3). (See 'Bradycardia algorithm' below.)

Bradycardia is defined as a heart rate that is slow compared with normal heart rates for the patient's age (table 6) [14].

Primary bradycardia is the result of congenital and acquired heart conditions that directly slow the spontaneous depolarization rate of the heart's pacemaker or slow conduction through the heart's conduction system.

Secondary bradycardia is the result of conditions that alter the normal function of the heart, including hypoxia, acidosis, hypotension, hypothermia, and drug effects.

The clinical presentation and the approach to determining the type of rhythm disturbance (sinus bradycardia versus atrioventricular [AV] block) are discussed in detail separately. (See "Bradycardia in children", section on 'Clinical evaluation' and "Bradycardia in children", section on 'Identifying the type and cause of bradycardia'.)

Tachycardia — Relative tachycardia is a heart rate that is too fast for the child's age, level of activity, and clinical condition (table 6). In children, sinus tachycardia usually represents hypovolemia, fever, physiologic response to stress, pain, or fear, or drug effect (such as with beta agonists). (See "Approach to the child with tachycardia".)

Pediatric respiratory rate and heart rate lower limit, normal range, and upper limit by age*

Table 6 - larger image below

Tachyarrhythmias are fast abnormal rhythms originating in the atria or the ventricles. Certain arrhythmias, such as supraventricular tachycardia and ventricular tachycardia, can lead to shock and cardiac arrest. Unstable rhythms lead to poor tissue perfusion with a fall in cardiac output, poor coronary artery perfusion, and increased myocardial oxygen demand, which can all lead to cardiogenic shock.

Signs and symptoms in children with tachycardia are often nonspecific and vary by age. They may include palpitations, lightheadedness, dizziness, fatigue and syncope. In infants, prolonged tachycardia may cause poor feeding, tachypnea, and irritability with signs of heart failure. (See "Approach to the child with palpitations" and "Emergency evaluation of syncope in children and adolescents".)

Important ECG findings include:

- Heart rate that is fast compared with normal rates (table 6)
- P waves that may or may not be visible
- QRS interval that is narrow or wide

Treatment priorities in managing tachycardias rely on whether hemodynamic instability is present and differentiating between tachycardia with narrow QRS complex (sinus tachycardia, supraventricular tachycardia, atrial flutter) and wide QRS complex tachycardias (ventricular tachycardia, supraventricular tachycardia with aberrant intraventricular conduction) (algorithm 4):

Pediatric tachycardia with a pulse 2020 update

Algorithm 4 - larger image below

- **Sinus tachycardia** – Sinus tachycardia is characterized by a rate of sinus node discharge that is faster than normal for the patient's age (table 6). This rhythm usually represents the body's increased need for cardiac output or oxygen delivery. The heart rate is not fixed and varies with other factors, including fever, stress, and level of activity. Causes include tissue hypoxia, hypovolemia, fever, metabolic stress, injury, pain, anxiety, toxins/poisons/drugs, and anemia. Less common causes include cardiac tamponade, tension pneumothorax, and thromboembolism. (See "Approach to the child with tachycardia".)

  Typical ECG findings in patients with sinus tachycardia include [14]:

  - Heart rate is usually <220/min in infants, <180/min in children, and exhibits beat to beat variability in rate.
  - P waves are present with normal appearance.
  - PR interval is constant and exhibits a normal duration for age.
  - R-R interval is variable.
  - QRS complex is narrow.

- **Supraventricular tachycardia** – Supraventricular tachycardia (SVT) can be defined as an abnormally rapid heart rhythm originating above the ventricles, often (but not always) with a narrow QRS complex; it conventionally excludes atrial flutter and atrial fibrillation. The two most common forms of SVT in children are atrioventricular reentrant tachycardia (AVRT), including the Wolff-Parkinson-White (WPW) syndrome (waveform 1), and atrioventricular nodal reentrant tachycardia (AVNRT).

  - **Signs and symptoms** – SVT typically has an abrupt onset and intermittent presentation. Signs and symptoms in infants include poor feeding, tachypnea, irritability, increased sleepiness, diaphoresis, pallor, and/or vomiting. Older children may have palpitations, shortness of breath, chest pain/discomfort, dizziness, lightheadedness, and/or fainting. Infants and children with prolonged SVT may display clinical findings of heart failure. (See "Clinical features and diagnosis of supraventricular tachycardia (SVT) in children", section on 'Clinical features'.)

    Typical ECG findings in patients with SVT include [14]:

    - Heart rate that is usually >220/min in infants, >180/min in children, and has **no** or minimal beat to beat variability.
    - P waves are absent or abnormal.
    - PR interval may not be present or short PR interval with ectopic atrial tachycardia.
    - R-R interval is usually constant.
    - QRS is usually narrow. Conduction delay along the ventricular system may lead to an appearance of wide complex tachycardia, known as SVT with aberrant conduction.

- **Ventricular tachycardia** – Ventricular tachycardia (VT) originates from the ventricular myocardium or Purkinje cells below the bifurcation of the bundle of His (waveform 2). VT may present with or without pulses. VT is associated with sudden cardiac death. As a result, patients who develop VT or are at risk for developing VT must be identified, evaluated, and treated, if necessary.

  VT with pulses can vary in rate from near normal to >200 beats per minute. Faster rates can compromise stroke volume and cardiac output leading to pulseless VT or ventricular fibrillation (VF). Causes of VT include underlying heart disease or cardiac surgery, prolonged QT syndrome or other channelopathies, or myocarditis/cardiomyopathy. Other causes include hyperkalemia and toxic ingestions (eg, tricyclic antidepressants, cocaine) (table 7).

  Findings of ventricular tachycardia on ECG include (waveform 2):

  - Ventricular rate is >120 beats per minute and regular.
  - P waves are often not identifiable, may have AV dissociation, or may have retrograde depolarization.
  - QRS is typically wide (>0.09 seconds).
  - T waves are often opposite in polarity from the QRS complex.

  Ventricular fibrillation, causes of wide complex QRS, and treatment of pulseless arrest are discussed separately. (See 'Pulseless arrest' below and "Causes of wide QRS complex tachycardia in children", section on 'Ventricular tachycardia' and 'Pulseless arrest algorithm' below.)

Pulseless arrest — Pulseless arrest refers to the cessation of blood circulation caused by absent or ineffective cardiac mechanical activity. Most pediatric cardiac arrests are hypoxic/asphyxial arrests that result from a progression of respiratory distress, respiratory failure, or shock rather than from primary cardiac arrhythmias ("sudden cardiac arrest").

Children with pulseless arrest appear apneic or display a few agonal gasps. They have no palpable pulses and are unresponsive.

The arrest rhythms consist of:

- **Shockable rhythms**:

  - **Ventricular fibrillation** – Ventricular fibrillation is characterized by no organized rhythm and no coordinated contractions (waveform 3). Electrical activity is chaotic. Causes overlap with etiologies of ventricular tachycardia, including hyperkalemia, congenital or acquired heart disease, toxic exposures, electrical or lightning shocks, and submersion.

  - **Pulseless ventricular tachycardia** – Pulseless VT is a cardiac arrest of ventricular origin characterized by organized, wide QRS complexes (waveform 2). Any cause of VT with pulses can lead to pulseless VT. (See 'Tachycardia' above.)

- **Torsades de pointes** – Torsades de pointes or polymorphic VT displays a QRS complex that changes in polarity and amplitude, appearing to rotate around the ECG isoelectric line (translation: "twisting of the points") (waveform 4). This arrhythmia is associated with markedly prolonged QTc interval from congenital conditions (long QT syndrome), drug toxicity (antiarrhythmic drugs, tricyclic antidepressants, calcium channel blockers, phenothiazine), and electrolyte disturbances (eg, hypomagnesemia arising from anorexia nervosa). Ventricular tachycardia, including torsades de pointes, can deteriorate into ventricular fibrillation.

- **Asystole** – Children with asystole have cardiac standstill with no discernible electrical activity (waveform 3). The most common cause is respiratory failure progressing to critical hypoxemia, bradycardia, and then cardiac standstill. Underlying conditions include airway obstruction, pneumonia, submersion, hypothermia, sepsis, and poisoning (eg, carbon monoxide poisoning, sedative-hypnotics) leading to hypoxia and acidosis.

- **Pulseless electrical activity** – Pulseless electrical activity (PEA) consists of any organized electrical activity observed on ECG in a patient with no central palpable pulse. Reversible conditions may underlie PEA, including:

  - Hypovolemia
  - Hypoxia
  - Hydrogen ion (acidosis)
  - Hypo-/hyperkalemia
  - Hypoglycemia
  - Hypothermia
  - Toxins
  - Tamponade, cardiac
  - Tension pneumothorax
  - Thrombosis (coronary or pulmonary)
  - Trauma

  These can be remembered as the H's and T's of PEA [15].

HIGHLY TRANSMISSIBLE AND VIRULENT INFECTIONS

The American Heart Association (AHA) in collaboration with several other major medical organizations has published guidance, including updated algorithms for basic and advanced life support for children with suspected or confirmed COVID-19 [16] with additional focused, pediatric-specific guidance from the American Academy of Pediatrics [17]. This guidance is also appropriate for the resuscitation of a pediatric patient suspected to have other highly transmissible and virulent infections.

For in-hospital pediatric cardiac arrest, modifications to advanced life support include (algorithm 5) [17]:

- Locate patients at risk for cardiac arrest in negative pressure rooms whenever possible.

- Don personal protective equipment (PPE) (eg, gown, N95 face mask, tight-fitting eye goggles or face shield, gloves) prior to entering the patient room and ensure that all members of the resuscitation team have the necessary PPE or immediately excuse them and replace them with properly protected providers.

- Keep the door to the resuscitation room closed.

- Limit personnel who are performing cardiopulmonary resuscitation (CPR). In general, a minimum of four providers is needed for high-quality pediatric advanced resuscitation. Video monitors, if available, can be used so that one team member may remain outside the room to monitor and record the resuscitation.

- Securely attach a high-efficiency particulate air (HEPA) filter with a low dead space to any ventilation device:

  - For unintubated patients:

    - Rapidly begin ventilating with a bag-mask and HEPA filter with a tight seal, ideally with a two-person technique (picture 1). Consider use of a supraglottic airway to optimize chest compression fraction prior to intubation.

    - Determine the need for endotracheal intubation based upon the overall clinical picture and per usual care standards.

- If performed, ensure endotracheal intubation with a cuffed endotracheal tube by the person most likely to achieve success on the first pass using a video laryngoscope, if available.
- Pause chest compression to perform endotracheal intubation.
- Connect the patient to a ventilator with inline HEPA filter and minimize any subsequent ventilator disconnections.

- For patients intubated and ventilated before arrest:
  - To reduce aerosol exposure, ensure that the endotracheal tube is connected, patent, and in proper position, and leave the patient connected to a closed ventilator system with an in-line HEPA filter.
  - For patients in the prone position at the time of arrest, initiate compressions with the hand centered over the T7 to T10 vertebral bodies, arrange sufficient numbers of trained personnel in PPE to safely turn the patient to the supine position, and then reinitiate CPR in the supine position.

- Ventilator settings during pediatric CPR:
  - Increase FiO_2 to 1 (100 percent inspired oxygen).
  - Based upon patient age and size, use either volume-controlled (table 8) or pressure-controlled (table 9) ventilator settings that produce adequate chest wall rise.
  - Initially set positive end-expiratory pressure (PEEP) to physiologic (3 to 5 cmH_2O); adjust as needed to optimize lung volume while maintaining adequate venous return.
  - Adjust trigger settings so that chest compressions do not cause the ventilator to initiate a breath.
  - Set the ventilation rate to 20 to 30 breaths per minute in infants and children.

Health care providers can significantly reduce their risk of infection, severe illness, and death by receiving recommended vaccines and boosters against pathogens to which they are likely to be exposed.

AHA RESUSCITATION GUIDELINES

The approach to pediatric advanced life support presented in this topic is consistent with the 2025 American Heart Association (AHA) and the American Academy of Pediatrics Guidelines on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care [1,2].

Bradycardia algorithm — The management of bradycardia in children should focus on optimizing oxygenation and ventilation and supporting circulation with chest compressions for patients with poor perfusion and a heart rate <60 beats per minute as shown in the algorithm (algorithm 3) and discussed separately. (See "Bradycardia in children", section on 'Patients with hemodynamic compromise'.)

Tachycardia algorithm — The management of sinus tachycardia focuses on treatment of the underlying physiologic derangement and is largely supportive.

The management of tachyarrhythmias that are not sinus in origin is guided by the appearance of the QRS complex, and by the patient's status, whether unstable or stable (algorithm 4) [1,2]:

- **Unstable** – Patients with a pulse and either narrow or wide complex tachycardia who have significantly impaired consciousness and hypotensive shock should be treated with synchronized cardioversion (initial dose: 0.5 to 1 J/kg) (algorithm 4). (See "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Methods: Manual defibrillator use'.)

- **Stable** – For patients who are mentating and not hypotensive, treatment is determined by the QRS complex:
  - **Narrow QRS (≤0.09 seconds)** – For narrow complex tachycardia suggestive of supraventricular tachycardia (SVT), vagal maneuvers may be attempted while preparing for medication administration. Appropriate vagal maneuvers include application of ice to the face or, in a cooperative child, a Valsalva maneuver by bearing down or blowing into an occluded straw.

    The first recommended medication for SVT is adenosine, 0.1 mg/kg (maximum dose 6 mg) administered rapidly IV/IO and followed by a rapid saline flush (table 10). (See "Management of supraventricular tachycardia (SVT) in children", section on 'SVT refractory to vagal maneuvers'.)

- **Wide QRS (>0.09 seconds)** – If the wide-complex rhythm is monomorphic and regular, it is acceptable to administer a dose of adenosine to determine if the rhythm is actually supraventricular tachycardia with aberrant conduction.

  Antiarrhythmic therapy of wide-complex tachycardia involves agents with significant side effects (eg, amiodarone or procainamide) and consultation with a pediatric cardiology specialist is strongly encouraged prior to administration. (See "Management and evaluation of wide QRS complex tachycardia in children", section on 'Acute management of stable patients'.)

Pulseless arrest algorithm — Treatment of a pediatric cardiac arrest is provided in the algorithm (algorithm 6) and summarized below [1,2]. The epidemiology of cardiac arrest in children is discussed separately. (See "Pediatric basic life support (BLS) for health care providers", section on 'Epidemiology and survival'.)

Pediatric cardiac arrest algorithm 2020 update

Algorithm 6 - larger image below

Start CPR — The first step is to initiate cardiopulmonary resuscitation according to the algorithms for a single rescuer or two or more rescuers (see "Pediatric basic life support (BLS) for health care providers"):

- Start compressions (see "Pediatric basic life support (BLS) for health care providers", section on 'Chest compressions')
- Begin bag-mask ventilation and give oxygen (see "Basic airway management in children", section on 'Bag-mask ventilation')
- Apply monitors/defibrillator (see "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Procedure')

Shockable rhythm — Patients with ventricular fibrillation (VF) or pulseless ventricular tachycardia (pVT) should receive immediate CPR and defibrillation at 2 J/kg of biphasic or monophasic energy as soon as a device is available (algorithm 6) [1,2]. After delivering the shock, perform approximately two minutes of CPR (10 cycles for two person CPR or 5 cycles for one person CPR) **before** checking the rhythm.

If the rhythm has not converted with defibrillation, then the patient should receive repeated defibrillation at a higher dose (4 J/kg) followed by additional cycles of CPR [1,2]. Subsequent defibrillations should be provided at a minimum of 4 J/kg, up to 10 J/kg or the adult energy dose (typically 120 to 200 J for a biphasic defibrillator and 360 J for a monophasic defibrillator).

Although manual defibrillators operated by advanced life support providers or automated external defibrillators with pediatric attenuating devices are preferred for use in infants and children, automated external defibrillators without pediatric attenuating devices may be used if they are the only option available. (See "Pediatric basic life support (BLS) for health care providers", section on 'Automated external defibrillator' and "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Methods: Manual defibrillator use'.)

Persistent VF or pVT requires the addition of medications such as parenteral epinephrine every three to five minutes and antiarrhythmic therapy (eg, amiodarone or lidocaine for VF or pVT (algorithm 6) or magnesium sulfate for torsades de pointes) [1,2,18]. When giving medications, the IO or IV route is always preferred to administration through the endotracheal tube. Attempts at vascular or intraosseous access should **not** interrupt chest compressions. During CPR, intraosseous access may be pursued initially, or simultaneously with peripheral vascular access. (See "Intraosseous infusion" and "Emergency and elective venous access in children", section on 'Emergency venous access'.)

Drug doses are as follows (table 10) [1,2,15]:

- **Epinephrine** – The IV/IO dose of epinephrine is 0.01 mg/kg (0.1 mL/kg of the 0.1 mg/mL concentration) given every three to five minutes; maximum single dose: 1 mg (10 mL).

  When epinephrine is administered via endotracheal tube, use a 10-fold higher dose or 0.1 mg/kg (0.1 mL/kg of the 1 mg/mL concentration) every three to five minutes. (See "Primary drugs in pediatric resuscitation", section on 'Epinephrine'.)

- **Lidocaine** – The initial IV/IO bolus dose of lidocaine is 1 mg/kg. This may be followed by an infusion of 20 to 50 mcg/kg/min. The bolus dose should be repeated if the lidocaine infusion is started more than 15 minutes after the initial bolus. Although lidocaine can be given through the endotracheal tube, the optimal dose is unknown. An increase of two- to threefold is suggested. (See "Primary drugs in pediatric resuscitation", section on 'Lidocaine' and "Primary drugs in pediatric resuscitation", section on 'Endotracheal drug administration'.)

- **Amiodarone** – The initial IV/IO dose of amiodarone is 5 mg/kg (maximum single dose 300 mg). If there is no clinical improvement, repeat amiodarone (5 mg/kg with maximum single dose 150 mg for subsequent doses) up to three times during acute treatment. (See "Primary drugs in pediatric resuscitation", section on 'Amiodarone'.)

- **Magnesium sulfate** – The IV/IO dose is 25 to 50 mg/kg (maximum dose 2 g) given as an infusion of magnesium sulfate diluted in a 5 percent dextrose solution (D5W) to a concentration of 20 percent or less and, in an arrested patient, infused over one to two minutes. (See "Primary drugs in pediatric resuscitation", section on 'Magnesium sulfate'.)

Resuscitation medications given through an IO or peripheral IV should be followed with a 5 to 10 mL flush of normal saline to move the drug from the peripheral to the central circulation.

Asystole or pulseless electrical activity — Patients with asystole or pulseless electrical activity should receive cardiopulmonary resuscitation and epinephrine as soon as possible after arrest [1,2].

During the course of the resuscitation, the clinician should evaluate for and treat underlying causes (H's and T's) for the pulseless arrest [2]. When giving medications, the IO or IV route is always preferred to administration through the endotracheal tube [2]. Attempts at vascular or intraosseous access should NOT interrupt chest compressions. During CPR, intraosseous access may be pursued initially, or simultaneously with peripheral vascular access. (See "Emergency and elective venous access in children", section on 'Emergency venous access'.)

The IV/IO dose of epinephrine is 0.01 mg/kg (0.1 mL/kg of the 0.1 mg/mL concentration) given every three to five minutes; maximum single dose: 1 mg (10 mL). Epinephrine given through an IO or peripheral IV should be followed with a 5 to 10 mL flush of normal saline to move the drug from the peripheral to the central circulation. IV/IO administration is strongly preferred to endotracheal (ET) administration.

When epinephrine is administered via ET tube, the dose should be increased 10-fold to 0.1 mg/kg (0.1 mL/kg of the 1 mg/mL concentration) every three to five minutes. (See "Primary drugs in pediatric resuscitation", section on 'Epinephrine' and "Primary drugs in pediatric resuscitation", section on 'Endotracheal drug administration'.)

Among children who arrest in an inpatient setting and who do not have rapid return of spontaneous circulation with initiation of basic life support, timely administration of epinephrine is associated with improved survival [19,20]. As an example, in a retrospective review of registry data on 1558 children with inpatient arrest and a documented non-shockable initial rhythm, the median time to the first dose of epinephrine was one minute [19,21]. Adjusted survival to discharge was seen in up to 37 percent of patients receiving epinephrine one minute or less after arrest and decreased 5 percent for every additional minute delay in epinephrine administration. Survival with favorable neurologic outcome at discharge occurred in approximately 16 percent of patients and also decreased 5 percent for every additional minute of delay in epinephrine administration based upon adjusted analysis.

Advanced airway placement — For children undergoing CPR in the out-of-hospital setting, bag-mask ventilation by prehospital providers is suggested rather than endotracheal intubation or supraglottic airway placement. (See "Pediatric considerations in prehospital care", section on 'Advanced airway management'.)

For children undergoing CPR in the inpatient hospital setting, the resuscitation team leader should determine the best time for advanced airway placement and ensure that interruption of cardiac compressions is minimized [22].

Once the trachea is intubated, chest compressions and ventilations can be performed independently. For infants and children, chest compressions are delivered at a rate of 100 to 120 per minute without pauses, and ventilations are given at a rate of 20 to 30 breaths per minute (one breath every two to three seconds) [2,22]. This ventilation rate in infants and children with an advanced airway is based upon a multicenter observational study of 47 children with in-hospital cardiopulmonary arrest (IHCA) that found that ventilation rates of at least 30 breaths per minute in infants and 25 breaths per minute in children were associated with increased rates of return of spontaneous circulation (ROSC) and survival [23].

Point-of-care ultrasonography — Point-of-care ultrasound by skilled clinicians may identify potentially treatable conditions in children with cardiac arrest such as pericardial tamponade, tension pneumothorax, or pulmonary embolus in selected patients. When performed, interruption of cardiac compressions should be minimized [1].

Monitoring — Given the importance of high-quality chest compressions, techniques to measure and monitor CPR performance have been developed. The availability of infant or pediatric CPR quality feedback using electrode pads is variable between manufacturers but can be used if available [2]. (See "Pediatric basic life support (BLS) for health care providers", section on 'Chest compressions'.)

In addition to monitoring rate and clinical effectiveness of ventilation, we use $EtCO_2$ measurements from continuous waveform capnography whenever possible during pediatric cardiac arrest. Sudden, sustained increases in $EtCO_2$ during CPR are associated with a return of spontaneous circulation (ROSC). In pediatric patients with a declining $EtCO_2$, efforts to improve the quality of compressions and to avoid excessive ventilation are appropriate [1,2]. (See "Carbon dioxide monitoring (capnography)", section on 'Return of spontaneous circulation'.)

For patients with invasive arterial monitoring in place, clinicians should target a diastolic blood pressure ≥25 mmHg for infants <1 year and ≥30 mmHg for children one year of age and older [1,2]. Published evidence from analysis of hemodynamic monitoring data during in-hospital pediatric CPR has found an association between a mean diastolic blood pressure (DBP) of >25 mmHg in infants young than one year of age and >30 mmHg in children one year of age and older and survival to discharge, as well as survival to discharge with favorable neurologic outcome [24].

Extracorporeal membrane oxygenation (ECMO) with CPR (ECPR) — Use of ECPR in settings with existing ECMO protocols, expertise, and equipment may be beneficial for selected patients not responsive to conventional CPR after inpatient cardiac arrest [1,2,22,25]. Our approach is to activate ECPR after approximately 5 to 10 minutes of failed conventional resuscitation in hospitalized patients with conditions that may be reversible after a period of mechanical circulatory support (eg, myocarditis, pulmonary or air embolus, sudden arrest after cardiac surgery, poisoning, or primary hypothermic arrest) or who are candidates for the use of ECPR as a bridge to therapies such as cardiac transplantation.

Extracorporeal membrane oxygenation with CPR (ECPR) for infants and children with in-hospital cardiac arrests is used in selected institutions with resources to rapidly deploy ECMO when conventional resuscitation efforts are unsuccessful [1,2,26,27]. Evidence supporting ECPR after inpatient cardiac arrest in children comes from observational studies with favorable neurologic outcomes in 40 to 66 percent of patients who survive to discharge [27-31]. For such patients, intact survival approaching 50 percent has been described even after prolonged periods of chest compressions (>60 minutes) [30]. Improved outcomes compared with conventional CPR has been primarily seen in patients with underlying cardiac diseases (eg, cardiomyopathy, myocarditis, or congenital cardiac anomalies). For example, in a multicenter prospective cohort study of 3756 children with inpatient cardiac arrests, ECPR was associated with overall increased rates of survival to discharge (40 versus 27 percent, adjusted odds ratio [aOR] 2.8, 95% CI 2.1 to 2.7) and favorable neurologic outcomes (27 versus 18 percent, aOR 2.6, 95% CI 1.9 to 3.6) compared with conventional CPR [27]. However, the study pooled patients with cardiac and noncardiac etiologies even though ECPR was much more likely to be used in surgical cardiac patients. When these groups were analyzed separately, only the cardiac patients had statistically significant improved outcomes.

By contrast, the benefit of ECPR for out-of-hospital arrest is less clear. Emerging evidence suggests that while overall survival remains poor, carefully selected children with out-of-hospital cardiac arrest (OHCA) may benefit from ECPR. For example, in a retrospective study of 80 children with OCHA who underwent ECPR, 29 percent had survival to discharge [32]. Factors associated with an increased odds of survival included an initial shockable rhythm and signs of life prior to initiation of ECPR. Further study is needed to determine selection criteria for ECPR in children with OHCA and to provide data on long-term neurologic outcomes.

Termination of resuscitation — Although certain factors are associated with better or worse outcomes after cardiac arrest in infants and children, no single factor is reliable enough to accurately guide whether termination efforts should cease or continue [1,2].

Thus, the decision to terminate resuscitation should be individualized and multiple factors considered including:

- Duration of cardiac arrest, including when the patient was discovered relative to initial presentation (eg, patients with sudden infant death syndrome who are found with evidence of lividity would have CPR discontinued earlier than patients with in-hospital arrests)
- Presenting rhythm (eg, shockable versus asystole or pulseless electrical activity)
- Underlying disease or cause, if known (eg, cardiac disease, trauma, respiratory failure, or sepsis)
- Setting and available resources
- Do not resuscitate status

Intact survival after prolonged resuscitation (>30 minutes) has occurred in patients with the following conditions [15]:

- Poisoning

- Primary hypothermic arrest (see "Hypothermia in children: Management", section on 'Nonperfusing cardiac rhythms')
- Patients with cardiac disease resuscitated with ECPR (see 'Extracorporeal membrane oxygenation (ECMO) with CPR (ECPR)' above)

Family presence during resuscitation — Observational studies indicate that caretakers should be given the option of being present during the in-hospital resuscitation of their child [33].

Key findings include:

- Most parents or primary caregivers want the opportunity to remain with their child during resuscitation [33] and believe it is their right [34].
- Caretakers present during the resuscitation of a family member frequently reported that their presence during the resuscitation was beneficial to the patient [34-36].
- Two-thirds of caretakers present during the resuscitation of a child who died reported that their presence helped with their adjustment to the death and the grieving process [36].
- Studies of hospital personnel suggest that the presence of a family member, in most instances, was not stressful to staff and did not negatively impact staff performance [34,35,37].

When family members are present during a pediatric resuscitation, a staff member with clinical knowledge, empathy, and strong interpersonal skills should be present with them to provide support and answer questions.

Early post-cardiac arrest management — The early postresuscitation period involves the time soon after return of spontaneous circulation or recovery from circulatory or respiratory failure up to 12 hours post-event. During this time, the clinician must continue to treat the underlying cause of the life-threatening event and monitor for common respiratory or circulatory problems that may cause secondary morbidity or death. Initial post-cardiac arrest care is discussed separately. (See "Initial post-cardiac arrest care in children".)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Basic and advanced cardiac life support in children".)

SUMMARY AND RECOMMENDATIONS

- **Early recognition** – The principal aim for Pediatric Advanced Life Support (PALS) is to prevent cardiopulmonary failure and arrest through early recognition and management of respiratory distress, respiratory failure, and shock. For hospitals that care for children, implementation of a Pediatric Early Warning System (PEWS) and a rapid response team (RRT) is advised. (See 'Assessment' above and 'Prevention of cardiac arrest' above.)
- **Respiratory distress and failure** – A major goal of pediatric advanced life support is to recognize and treat respiratory conditions amenable to simple measures (eg, supplemental oxygen, inhaled albuterol) (table 2). The clinician may also have to treat rapidly progressive conditions and intervene with advanced therapies to avoid cardiopulmonary arrest in patients with respiratory failure. Early detection and treatment improve overall outcome. (See 'Respiratory distress and failure' above.)

  Key steps in basic airway management include (see 'Respiratory distress and failure' above):

  - Provide 100 percent inspired oxygen
  - Allow child to assume position of comfort or manually open airway
  - Clear airway (suction)
  - Insert an airway adjunct if consciousness is impaired (eg, nasopharyngeal airway or, if gag reflex absent, oropharyngeal airway)

  The clinician should assist ventilation manually in patients not responding to basic airway maneuvers, monitor oxygenation by pulse oximetry, monitor ventilation by end-tidal carbon dioxide ($EtCO_2$) if available, and administer medications as needed (eg, albuterol or racemic epinephrine). In preparation for intubation, the patient should receive 100 percent oxygen via a high-

concentration mask, or if indicated, positive pressure ventilation with a bag-valve-mask to preoxygenate and improve ventilation. (See 'Respiratory distress and failure' above.)

Children who cannot maintain an effective airway, oxygenation, or ventilation should receive noninvasive ventilation (NIV) or undergo endotracheal intubation. A rapid overview provides the steps in performing rapid sequence intubation (table 3). Initiation of NIV is discussed separately. (See "Rapid sequence intubation (RSI) in children for emergency medicine: Approach" and "Noninvasive ventilation for acute and impending respiratory failure in children".)

- **Shock** – Proper treatment of shock in children requires the clinician to recognize and eventually categorize the type of shock in order to prioritize treatment options (algorithm 1). Early treatment of shock may prevent the progression to cardiopulmonary failure (algorithm 2). (See "Initial evaluation of shock in children" and "Shock in children in resource-abundant settings: Initial management" and 'Shock' above.)

- **Bradycardia** – The initial management of bradycardia is provided in the algorithm (algorithm 3). The clinical presentation and the approach to determining the type of rhythm disturbance (sinus bradycardia versus atrioventricular [AV] block) are discussed in detail separately. (See "Bradycardia in children", section on 'Clinical evaluation' and "Bradycardia in children", section on 'Identifying the type and cause of bradycardia'.)

- **Tachycardia** – The management of tachyarrhythmias that are not sinus in origin is guided by the appearance of the QRS complex and by the patient's status, whether unstable or stable (algorithm 4). (See 'Tachycardia algorithm' above.)

- **Pulseless arrest** – Resuscitation of children in cardiopulmonary arrest is determined by the presenting rhythm (shockable versus pulseless electrical activity or asystole) (algorithm 6). (See 'Pulseless arrest algorithm' above.)

  Use of extracorporeal membrane oxygenation (ECMO) with CPR (ECPR) in settings with existing ECMO protocols, expertise, and equipment may be beneficial for selected patients who fail conventional CPR after inpatient cardiac arrest. Our approach is to activate ECPR after approximately 10 minutes of failed conventional resuscitation in patients with conditions that may be reversible after a period of ECPR (eg, myocarditis, pulmonary or air embolus, sudden arrest after cardiac surgery, poisoning, or primary hypothermic arrest) or who are candidates for the use of ECPR as a bridge to therapies such as cardiac transplantation. (See 'Extracorporeal membrane oxygenation (ECMO) with CPR (ECPR)' above.)

- **Postresuscitation management** – Initial post-cardiac arrest care is discussed separately. (See "Initial post-cardiac arrest care in children".)

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Topic 6392 Version 72.0

GRAPHICS

Table 1: **Glasgow Coma Scale and Pediatric Glasgow Coma Scale**

Sign	Glasgow Coma Scale[1]	Pediatric Glasgow Coma Scale[2]	Score
Eye opening	Spontaneous	Spontaneous	4
	To command	To sound	3
	To pain	To pain	2
	None	None	1
Verbal response	Oriented	Age-appropriate vocalization, smile, or orientation to sound; interacts (coos, babbles); follows objects	5
	Confused, disoriented	Cries, irritable	4
	Inappropriate words	Cries to pain	3
	Incomprehensible sounds	Moans to pain	2
	None	None	1
Motor response	Obeys commands	Spontaneous movements (obeys verbal command)	6
	Localizes pain	Withdraws to touch (localizes pain)	5
	Withdraws	Withdraws to pain	4
	Abnormal flexion to pain	Abnormal flexion to pain (decorticate posture)	3
	Abnormal extension to pain	Abnormal extension to pain (decerebrate posture)	2
	None	None	1
Best total score			**15**

The Glasgow Coma Scale (GCS) is scored between 3 and 15, with 3 being the worst and 15 the best. It is composed of 3 parameters: best eye response (E), best verbal response (V), and best motor response (M). The components of the GCS should be recorded individually; for example, E2V3M4 results in a GCS of 9. Traditionally, the GCS defines the severity of traumatic brain injury (TBI) as follows: ≤8: severe brain injury, 9 to 12: moderate injury, and a score ≥13 or higher: mild injury. However, a significant minority of patients with TBI and a GCS score of 13 have potentially life-threatening intracranial lesions. While a revised classification has not been widely adopted, a GCS score of 9 through 13 likely best represents the TBI population at moderate risk for death or long-term disability (ie, "potentially severe").

The Pediatric Glasgow Coma Scale (PGCS) was validated in children 2 years of age or younger.

Data from:

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Graphic 59662 Version 15.0

Table 2: **Causes of acute respiratory distress in children**

Respiratory tract
Infection
Uvulitis
Epiglottitis*
Retropharyngeal abscess
Peritonsillar abscess
Croup
Tracheitis
Bronchiolitis
Pneumonia¶
Asthma
Anaphylaxis*
Foreign body (upper airway*, lower airway, esophagus)
Airway anomalies (eg, laryngomalacia, laryngospasm, tracheoesophageal fistula, tracheal stenosis, tracheal ring or sling)
Biologic or chemical weapons* (eg, anthrax, tularemia, phosgene, nitrogen mustard, nerve agents, ricin)
Chest wall trauma or abnormalities (eg, flail chest*, open pneumothorax*, thoracic dystrophy)
Thoracic cavity trauma or conditions (eg, pneumothorax*, hemothorax*, pleural effusion, empyema, mediastinal mass)
Pulmonary trauma or conditions (contusion, embolism, hemorrhage)
Smoke inhalation*
Chemical agent exposures* (eg, phosgene, chlorine, cyanide)
Submersion injury (near-drowning)*
Cardiovascular
Congenital heart disease*
Acute decompensated heart failure*
Myocarditis*
Pericarditis
Arrhythmia*
Shock*
Cardiac tamponade*
Myocardial infarction*
Nervous system
Depressed ventilation* (eg, ingestion, CNS trauma, seizures, or CNS infection)
Hypotonia* (conditions causing poor airway or respiratory muscle tone and ineffective respiratory effort)
Pulmonary aspiration due to loss of airway protective reflexes
Gastrointestinal
Hypoventilation due to abdominal pain or distention (eg, intraabdominal trauma, small bowel obstruction, bowel perforation)
Gastroesophageal reflux with pulmonary aspiration
Metabolic and endocrine diseases
Metabolic acidosis (eg, diabetic ketoacidosis, severe dehydration, sepsis, toxic ingestions, inborn errors of metabolism)
Hyperthyroidism
Hypothyroidism
Hyperammonemia
Hypocalcemia (laryngospasm)
Hematologic

Decreased O_2 carrying capacity (eg, acute severe anemia from blood loss or hemolysis, hemoglobin variants, methemoglobinemia, or carbon monoxide poisoning)
Acute chest syndrome (patients with sickle cell disease)*

CNS: central nervous system; O_2: oxygen.
*Condition that can be immediately life threatening.
¶ Pneumonia may be caused by a variety of bacterial and viral pathogens. COVID-19 should also be considered in acutely ill children presenting with pneumonia in regions with a high prevalence although severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) rarely causes serious illness in the pediatric age group. Refer to UpToDate topics on COVID-19 infection in children.

Graphic 61637 Version 14.0

Table 3: **Rapid sequence intubation in children: Rapid overview of emergency management**

Preparation – Utilize an active checklist to:
Begin preoxygenation as described below.
Identify conditions that will affect choice of medications (eg, increased intracranial pressure, septic shock, bronchospasm, status epilepticus, or, if succinylcholine use is planned, absolute contraindications for its use as listed below).
Identify conditions that will predict difficult intubation or bag-mask ventilation (eg, small chin, inability to fully open the mouth, upper airway trauma, or infection).
Assemble equipment and check for function.
Develop contingency plan for failed intubation (refer to UpToDate topics on devices for difficult endotracheal intubation).
Preoxygenation
Begin preoxygenation as soon as rapid sequence intubation is potentially needed: ▪ Spontaneously breathing: 100% FiO_2 (>7 L/min oxygen flow) by nonrebreather mask for 3 minutes ▪ Apneic or inadequate breathing: Bag-mask ventilation with small tidal breaths using 100% FiO_2 ▪ During induction and paralysis, provide apneic oxygenation via nasal cannula at a flow rate of 1 L/min per year of age (minimum flow 2 L/min and maximum flow 15 L/min)
Administer oxygen at the highest concentration available.
Pretreatment (optional)
Atropine: Although not routinely recommended, many experts suggest atropine as pretreatment for: ▪ Children ≤1 year ▪ Children in shock ▪ Children <5 years receiving succinylcholine ▪ Older children receiving a second dose of succinylcholine Dose: 0.02 mg/kg IV without a minimum dose (maximum single dose 1 mg; if no IV access, can be given IM).
Induction (sedation)
Etomidate: ▪ Safe in patients with hemodynamic instability and is neuroprotective; an option in patients with sepsis and septic shock if ketamine is otherwise contraindicated or not readily available but causes transient adrenal cortico-suppression. ▪ Dose: 0.3 mg/kg IV.
Ketamine: ▪ Safe in patients with hemodynamic instability if patient is not catecholamine depleted. Generally preferred in patients with bronchospasm and sepsis. Use with **caution** in hypertensive patients with increased intracranial pressure. ▪ Dose: 1 to 2 mg/kg IV (if no IV access, can be given IM dose: 3 to 7 mg/kg).
Propofol: ▪ Causes hypotension. May use in hemodynamically stable patients with status epilepticus. ▪ Dose 1 to 1.5 mg/kg IV.
Midazolam: ▪ May use in hemodynamically stable patients with status epilepticus. Time to clinical effect is longer; inconsistently induces unconsciousness. May cause hemodynamic instability at doses required for sedation. ▪ Dose: 0.2 to 0.3 mg/kg IV (maximum dose 10 mg; onset of effect requires 2 to 3 minutes).
Fentanyl: ▪ Optional for cardiogenic shock or catecholamine-depleted shock (eg, persistent hypotension despite vasopressor therapy). Limited evidence in children. ▪ Dose 1 to 5 mcg/kg titrated to effect. Start at lower end of range in hypotensive patients. Give over 30 to 60 seconds to avoid respiratory depression or chest wall rigidity.
Paralytic
Rocuronium: ▪ Use for children with contraindication for succinylcholine or as primary paralytic if sugammadex is immediately available. ▪ Dose: 1 mg/kg IV.*
Succinylcholine: ▪ Do not use with extensive crush injury with rhabdomyolysis, chronic skeletal muscle disease (eg, Becker muscular dystrophy) or denervating neuromuscular disease (eg, cerebral palsy with paralysis); 48 to 72 hours after burn, multiple trauma, or denervating injury; patients with history or malignant hyperthermia; or pre-existing hyperkalemia.

- Dose: Infants and children ≤2 years: 2 mg/kg IV, older children and adolescents: 1 to 1.5 mg/kg IV (if IV access unobtainable, can be given IM, dose: 4 mg/kg)¶.

Protection and positioning
Maintain manual cervical spine immobilization during intubation in the trauma patient.
If cervical spine injury is not potentially present, put the patient in the "sniffing position" (ie, head forward so that the external auditory canal is anterior to the shoulder and the nose and mouth point to the ceiling).
Utilize external laryngeal manipulation or, in infants, gentle cricoid pressure to optimize the view of the glottis during direct laryngoscopy if the initial view is suboptimal or inadequate despite correct laryngoscope blade positioning.Δ
Positioning, with placement
Confirm tracheal tube placement with end-tidal CO_2 detection and auscultation.
Postintubation management
Obtain a chest radiograph to confirm the depth of tracheal tube insertion.
Provide ongoing sedation (eg, midazolam), analgesia (eg, fentanyl 1 mcg/kg), and, if indicated, paralysis.◇

If IV access is not rapidly obtained, intraosseous administration of drugs is an acceptable alternative.

CO_2: carbon dioxide; FiO_2: fraction of inspired oxygen; IM: intramuscularly; IV: intravenously.

* Sugammadex in a dose of 16 mg/kg can provide immediate reversal of paralysis when given approximately 3 minutes after a single dose of rocuronium or vecuronium. Vecuronium may be used in children with contraindications to succinylcholine and when rocuronium is not available. Suggested dose for rapid sequence intubation: vecuronium 0.15 to 0.2 mg/kg. Patients may experience prolonged and unpredictable duration of paralysis at this dose.

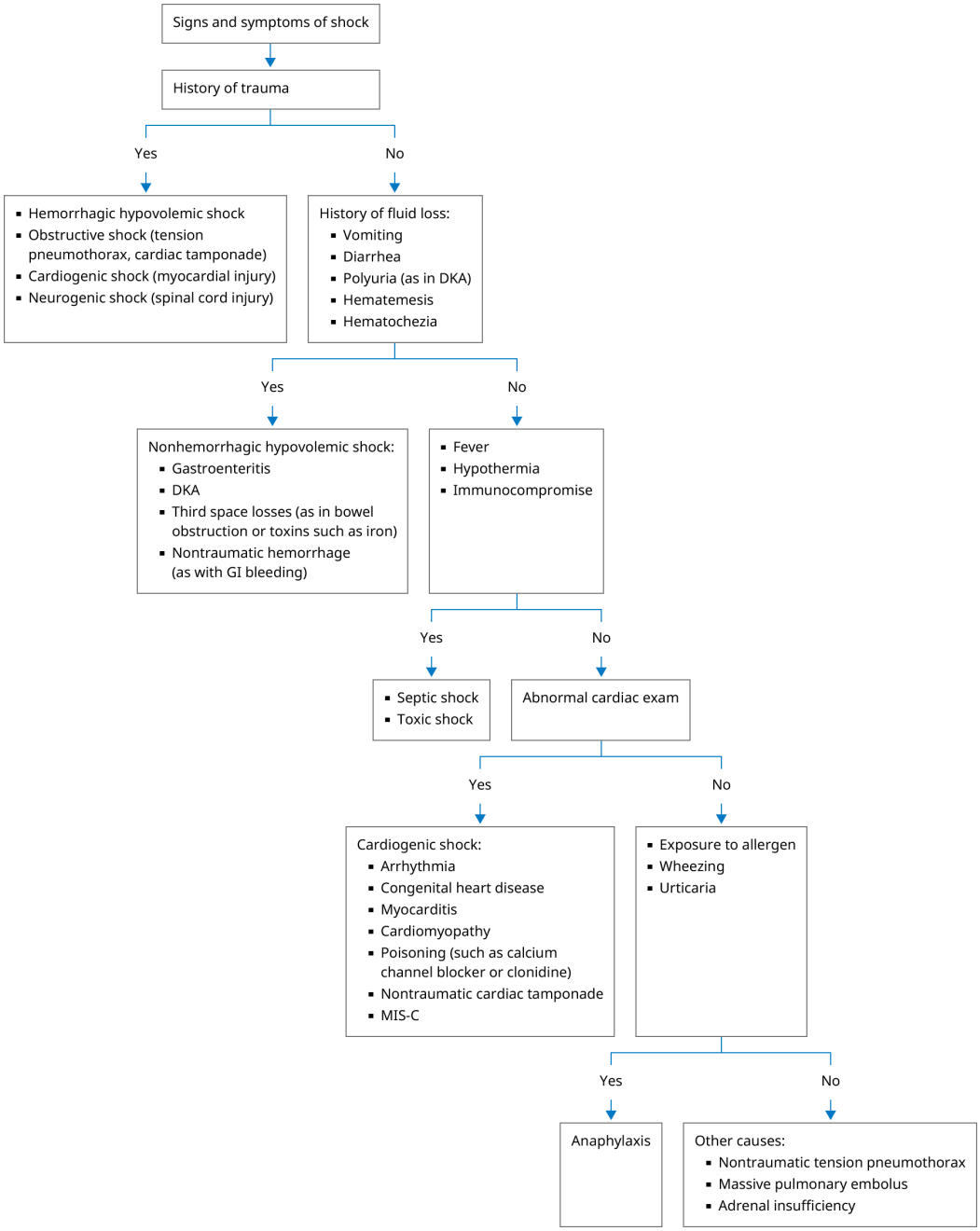
¶ Defasciculating agents (eg, rocuronium or vecuronium at one-tenth of the paralyzing dose) are not routinely recommended for children receiving succinylcholine. Onset of paralysis is slower by the IM route; the clinician must ensure full pre-oxygenation prior to administration, whenever possible, and be prepared to perform bag-mask ventilation if desaturation occurs before the patient is fully paralyzed for endotracheal intubation.

Δ Bimanual laryngoscopy, also called external laryngeal manipulation (ELM), entails manipulating the thyroid cartilage or hyoid bone with the right hand during laryngoscopy in order to improve the view of the glottis. For a description of how to perform ELM, refer to UpToDate topics on emergency endotracheal intubation in children and rapid sequence intubation in children.

◇ If decompensation occurs after successful intubation, use the DOPE mnemonic to find the cause:
- D: Dislodgement of the tube (right mainstem or esophageal)
- O: Obstruction of tube
- P: Pneumothorax
- E: Equipment failure (ventilator malfunction, oxygen disconnected or not on)

Graphic 51456 Version 46.0

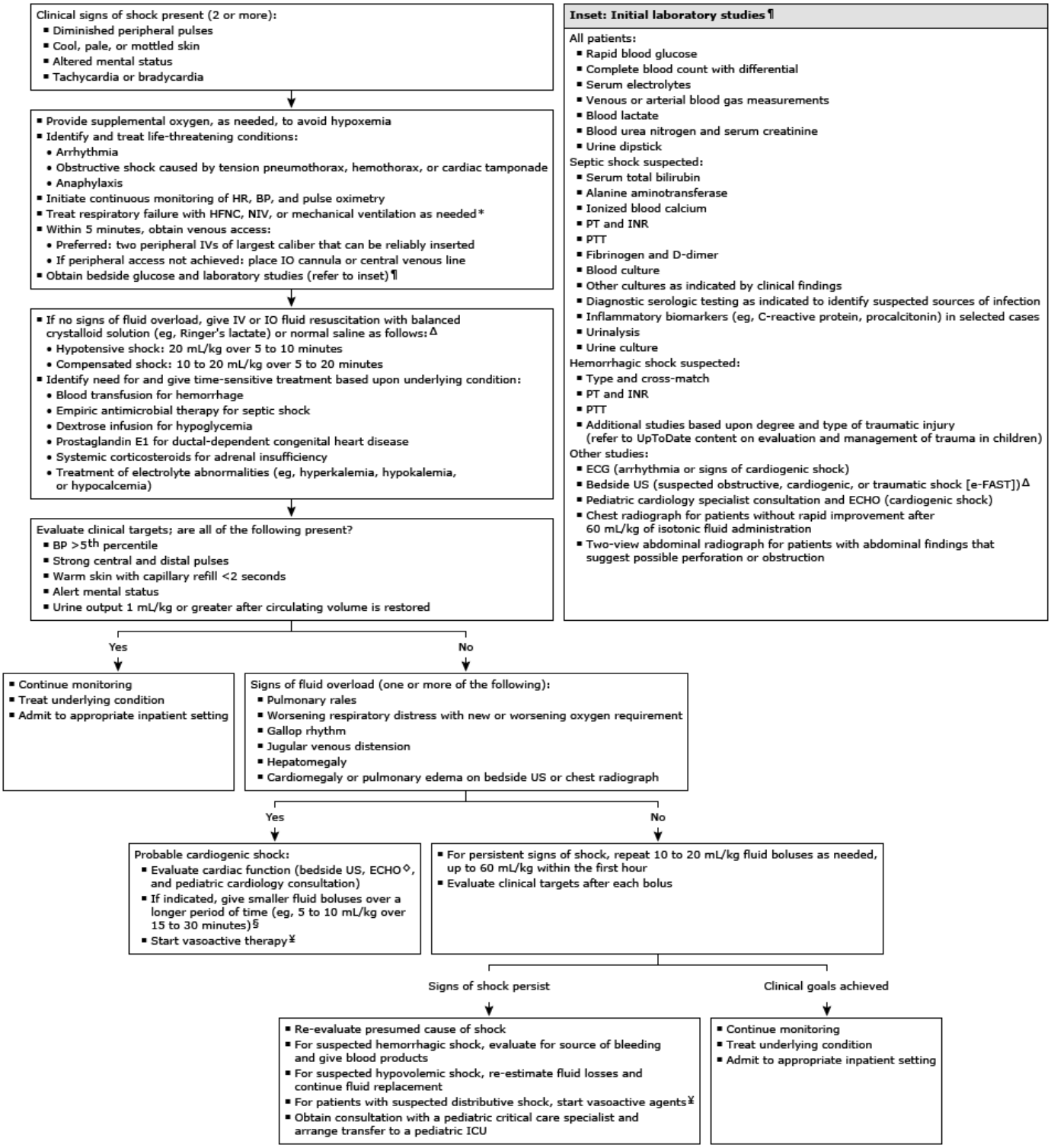
Algorithm 1: **Approach to the classification of undifferentiated shock in children**



DKA: diabetic ketoacidosis; GI: gastrointestinal; MIS-C: multisystem inflammatory syndrome in children.

Graphic 81356 Version 7.0

Algorithm 2: **Initial shock management in children in settings with access to ICU care**



ICU: intensive care unit; HR: heart rate; BP: blood pressure; HFNC: high-flow oxygen by nasal cannula; NIV: noninvasive ventilation; IV: intravenous; IO: intraosseous; US: ultrasound; ECHO: echocardiography; PT: prothrombin time; INR: international normalized ratio; PTT partial thromboplastin time; ECG: electrocardiography; e-FAST: extended focused assessment with sonography for trauma.

* A trial of HFNC or NIV, such as continuous positive airway pressure ventilation or bi-level positive airway pressure ventilation, may avoid the need for endotracheal intubation in selected patients. Patients with hemodynamic instability should receive appropriate interventions to treat shock prior to or during intubation. Refer to UpToDate content on HFNC, NIV, and rapid sequence intubation in children.

¶ Ancillary studies are determined by patient presentation and suspected type or types of shock present. Other laboratory and ancillary studies may also be indicated based upon the suspected underlying condition that is causing shock.

Δ Fluid volume should be calculated based upon ideal body weight (eg, 50th percentile for age).

◇ When performed by trained and experienced physicians, bedside ECHO can provide rapid evidence of myocardial dysfunction, including dysfunction due to obstructive shock.

§ Patients with signs of fluid overload who continue to receive fluid boluses warrant close monitoring for respiratory and cardiac failure. The clinician should have a low threshold for endotracheal intubation and mechanical ventilation to treat pulmonary edema in these patients.

¥ Suggested vasoactive therapy depends upon type of shock and clinical findings; refer to UpToDate topics and graphics on management of shock in children.

Graphic 129655 Version 1.0

Table 4: **Blood pressure levels for males by age and height percentile**

BP (percentile)	Systolic BP (mmHg)							Diastolic BP (mmHg)						
	Height percentile or measured height							Height percentile or measured height						
	5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
1 year														
Height (in)	30.4	30.8	31.6	32.4	33.3	34.1	34.6	30.4	30.8	31.6	32.4	33.3	34.1	34.6
Height (cm)	77.2	78.3	80.2	82.4	84.6	86.7	87.9	77.2	78.3	80.2	82.4	84.6	86.7	87.9
50th	85	85	86	86	87	88	88	40	40	40	41	41	42	42
90th	98	99	99	100	100	101	101	52	52	53	53	54	54	54
95th	102	102	103	103	104	105	105	54	54	55	55	56	57	57
95th + 12 mmHg	114	114	115	115	116	117	117	66	66	67	67	68	69	69
2 years														
Height (in)	33.9	34.4	35.3	36.3	37.3	38.2	38.8	33.9	34.4	35.3	36.3	37.3	38.2	38.8
Height (cm)	86.1	87.4	89.6	92.1	94.7	97.1	98.5	86.1	87.4	89.6	92.1	94.7	97.1	98.5
50th	87	87	88	89	89	90	91	43	43	44	44	45	46	46
90th	100	100	101	102	103	103	104	55	55	56	56	57	58	58
95th	104	105	105	106	107	107	108	57	58	58	59	60	61	61
95th + 12 mmHg	116	117	117	118	119	119	120	69	70	70	71	72	73	73
3 years														
Height (in)	36.4	37.0	37.9	39.0	40.1	41.1	41.7	36.4	37.0	37.9	39.0	40.1	41.1	41.7
Height (cm)	92.5	93.9	96.3	99.0	101.8	104.3	105.8	92.5	93.9	96.3	99.0	101.8	104.3	105.8
50th	88	89	89	90	91	92	92	45	46	46	47	48	49	49
90th	101	102	102	103	104	105	105	58	58	59	59	60	61	61
95th	106	106	107	107	108	109	109	60	61	61	62	63	64	64
95th + 12 mmHg	118	118	119	119	120	121	121	72	73	73	74	75	76	76
4 years														
Height (in)	38.8	39.4	40.5	41.7	42.9	43.9	44.5	38.8	39.4	40.5	41.7	42.9	43.9	44.5
Height (cm)	98.5	100.2	102.9	105.9	108.9	111.5	113.2	98.5	100.2	102.9	105.9	108.9	111.5	113.2
50th	90	90	91	92	93	94	94	48	49	49	50	51	52	52
90th	102	103	104	105	105	106	107	60	61	62	62	63	64	64
95th	107	107	108	108	109	110	110	63	64	65	66	67	67	68
95th + 12 mmHg	119	119	120	120	121	122	122	75	76	77	78	79	79	80
5 years														
Height (in)	41.1	41.8	43.0	44.3	45.5	46.7	47.4	41.1	41.8	43.0	44.3	45.5	46.7	47.4
Height (cm)	104.4	106.2	109.1	112.4	115.7	118.6	120.3	104.4	106.2	109.1	112.4	115.7	118.6	120.3
50th	91	92	93	94	95	96	96	51	51	52	53	54	55	55
90th	103	104	105	106	107	108	108	63	64	65	65	66	67	67
95th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
95th + 12 mmHg	119	120	121	121	122	123	124	78	79	80	81	82	82	83
6 years														
Height (in)	43.4	44.2	45.4	46.8	48.2	49.4	50.2	43.4	44.2	45.4	46.8	48.2	49.4	50.2
Height (cm)	110.3	112.2	115.3	118.9	122.4	125.6	127.5	110.3	112.2	115.3	118.9	122.4	125.6	127.5
50th	93	93	94	95	96	97	98	54	54	55	56	57	57	58
90th	105	105	106	107	109	110	110	66	66	67	68	68	69	69
95th	108	109	110	111	112	113	114	69	70	70	71	72	72	73

95th + 12 mmHg	120	121	122	123	124	125	126	81	82	82	83	84	84	85
7 years														
Height (in)	45.7	46.5	47.8	49.3	50.8	52.1	52.9	45.7	46.5	47.8	49.3	50.8	52.1	52.9
Height (cm)	116.1	118.0	121.4	125.1	128.9	132.4	134.5	116.1	118.0	121.4	125.1	128.9	132.4	134.5
50th	94	94	95	97	98	98	99	56	56	57	58	58	59	59
90th	106	107	108	109	110	111	111	68	68	69	70	70	71	71
95th	110	110	111	112	114	115	116	71	71	72	73	73	74	74
95th + 12 mmHg	122	122	123	124	126	127	128	83	83	84	85	85	86	86
8 years														
Height (in)	47.8	48.6	50.0	51.6	53.2	54.6	55.5	47.8	48.6	50.0	51.6	53.2	54.6	55.5
Height (cm)	121.4	123.5	127.0	131.0	135.1	138.8	141.0	121.4	123.5	127.0	131.0	135.1	138.8	141.0
50th	95	96	97	98	99	99	100	57	57	58	59	59	60	60
90th	107	108	109	110	111	112	112	69	70	70	71	72	72	73
95th	111	112	112	114	115	116	117	72	73	73	74	75	75	75
95th + 12 mmHg	123	124	124	126	127	128	129	84	85	85	86	87	87	87
9 years														
Height (in)	49.6	50.5	52.0	53.7	55.4	56.9	57.9	49.6	50.5	52.0	53.7	55.4	56.9	57.9
Height (cm)	126.0	128.3	132.1	136.3	140.7	144.7	147.1	126.0	128.3	132.1	136.3	140.7	144.7	147.1
50th	96	97	98	99	100	101	101	57	58	59	60	61	62	62
90th	107	108	109	110	112	113	114	70	71	72	73	74	74	74
95th	112	112	113	115	116	118	119	74	74	75	76	76	77	77
95th + 12 mmHg	124	124	125	127	128	130	131	86	86	87	88	88	89	89
10 years														
Height (in)	51.3	52.2	53.8	55.6	57.4	59.1	60.1	51.3	52.2	53.8	55.6	57.4	59.1	60.1
Height (cm)	130.2	132.7	136.7	141.3	145.9	150.1	152.7	130.2	132.7	136.7	141.3	145.9	150.1	152.7
50th	97	98	99	100	101	102	103	59	60	61	62	63	63	64
90th	108	109	111	112	113	115	116	72	73	74	74	75	75	76
95th	112	113	114	116	118	120	121	76	76	77	77	78	78	78
95th + 12 mmHg	124	125	126	128	130	132	133	88	88	89	89	90	90	90
11 years														
Height (in)	53.0	54.0	55.7	57.6	59.6	61.3	62.4	53.0	54.0	55.7	57.6	59.6	61.3	62.4
Height (cm)	134.7	137.3	141.5	146.4	151.3	155.8	158.6	134.7	137.3	141.5	146.4	151.3	155.8	158.6
50th	99	99	101	102	103	104	106	61	61	62	63	63	63	63
90th	110	111	112	114	116	117	118	74	74	75	75	75	76	76
95th	114	114	116	118	120	123	124	77	78	78	78	78	78	78
95th + 12 mmHg	126	126	128	130	132	135	136	89	90	90	90	90	90	90
12 years														
Height (in)	55.2	56.3	58.1	60.1	62.2	64.0	65.2	55.2	56.3	58.1	60.1	62.2	64.0	65.2
Height (cm)	140.3	143.0	147.5	152.7	157.9	162.6	165.5	140.3	143.0	147.5	152.7	157.9	162.6	165.5
50th	101	101	102	104	106	108	109	61	62	62	62	62	63	63
90th	113	114	115	117	119	121	122	75	75	75	75	75	76	76
95th	116	117	118	121	124	126	128	78	78	78	78	78	79	79
95th + 12 mmHg	128	129	130	133	136	138	140	90	90	90	90	90	91	91
13 years														
Height (in)	57.9	59.1	61.0	63.1	65.2	67.1	68.3	57.9	59.1	61.0	63.1	65.2	67.1	68.3
Height (cm)	147.0	150.0	154.9	160.3	165.7	170.5	173.4	147.0	150.0	154.9	160.3	165.7	170.5	173.4

50th	103	104	105	108	110	111	112	61	60	61	62	63	64	65
90th	115	116	118	121	124	126	126	74	74	74	75	76	77	77
95th	119	120	122	125	128	130	131	78	78	78	78	80	81	81
95th + 12 mmHg	131	132	134	137	140	142	143	90	90	90	90	92	93	93
14 years														
Height (in)	60.6	61.8	63.8	65.9	68.0	69.8	70.9	60.6	61.8	63.8	65.9	68.0	69.8	70.9
Height (cm)	153.8	156.9	162.0	167.5	172.7	177.4	180.1	153.8	156.9	162.0	167.5	172.7	177.4	180.1
50th	105	106	109	111	112	113	113	60	60	62	64	65	66	67
90th	119	120	123	126	127	128	129	74	74	75	77	78	79	80
95th	123	125	127	130	132	133	134	77	78	79	81	82	83	84
95th + 12 mmHg	135	137	139	142	144	145	146	89	90	91	93	94	95	96
15 years														
Height (in)	62.6	63.8	65.7	67.8	69.8	71.5	72.5	62.6	63.8	65.7	67.8	69.8	71.5	72.5
Height (cm)	159.0	162.0	166.9	172.2	177.2	181.6	184.2	159.0	162.0	166.9	172.2	177.2	181.6	184.2
50th	108	110	112	113	114	114	114	61	62	64	65	66	67	68
90th	123	124	126	128	129	130	130	75	76	78	79	80	81	81
95th	127	129	131	132	134	135	135	78	79	81	83	84	85	85
95th + 12 mmHg	139	141	143	144	146	147	147	90	91	93	95	96	97	97
16 years														
Height (in)	63.8	64.9	66.8	68.8	70.7	72.4	73.4	63.8	64.9	66.8	68.8	70.7	72.4	73.4
Height (cm)	162.1	165.0	169.6	174.6	179.5	183.8	186.4	162.1	165.0	169.6	174.6	179.5	183.8	186.4
50th	111	112	114	115	115	116	116	63	64	66	67	68	69	69
90th	126	127	128	129	131	131	132	77	78	79	80	81	82	82
95th	130	131	133	134	135	136	137	80	81	83	84	85	86	86
95th + 12 mmHg	142	143	145	146	147	148	149	92	93	95	96	97	98	98
17 years														
Height (in)	64.5	65.5	67.3	69.2	71.1	72.8	73.8	64.5	65.5	67.3	69.2	71.1	72.8	73.8
Height (cm)	163.8	166.5	170.9	175.8	180.7	184.9	187.5	163.8	166.5	170.9	175.8	180.7	184.9	187.5
50th	114	115	116	117	117	118	118	65	66	67	68	69	70	70
90th	128	129	130	131	132	133	134	78	79	80	81	82	82	83
95th	132	133	134	135	137	138	138	81	82	84	85	86	86	87
95th + 12 mmHg	144	145	146	147	149	150	150	93	94	96	97	98	98	99

The 50th, 90th, and 95th percentiles were derived by using quantile regression on the basis of normal-weight children (BMI <85th percentile). BP stages are defined as elevated BP ≥90th percentile but <95th percentile; stage 1 HTN: ≥95th percentile or 130/80 to 139/89 mmHg; and stage 2 HTN: ≥95th percentile + 12 mmHg or >140/90 mmHg.

BMI: body mass index; BP: blood pressure; HTN: hypertension.

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Graphic 63856 Version 21.0

Table 5: **Blood pressure levels for females by age and height percentile**

BP (percentile)	Systolic BP (mmHg)							Diastolic BP (mmHg)						
	Height percentile or measured height							Height percentile or measured height						
	5%	10%	25%	50%	75%	90%	95%	5%	10%	25%	50%	75%	90%	95%
1 year														
Height (in)	29.7	30.2	30.9	31.8	32.7	33.4	33.9	29.7	30.2	30.9	31.8	32.7	33.4	33.9
Height (cm)	75.4	76.6	78.6	80.8	83.0	84.9	86.1	75.4	76.6	78.6	80.8	83.0	84.9	86.1
50th	84	85	86	86	87	88	88	41	42	42	43	44	45	46
90th	98	99	99	100	101	102	102	54	55	56	56	57	58	58
95th	101	102	102	103	104	105	105	59	59	60	60	61	62	62
95th + 12 mmHg	113	114	114	115	116	117	117	71	71	72	72	73	74	74
2 years														
Height (in)	33.4	34.0	34.9	35.9	36.9	37.8	38.4	33.4	34.0	34.9	35.9	36.9	37.8	38.4
Height (cm)	84.9	86.3	88.6	91.1	93.7	96.0	97.4	84.9	86.3	88.6	91.1	93.7	96.0	97.4
50th	87	87	88	89	90	91	91	45	46	47	48	49	50	51
90th	101	101	102	103	104	105	106	58	58	59	60	61	62	62
95th	104	105	106	106	107	108	109	62	63	63	64	65	66	66
95th + 12 mmHg	116	117	118	118	119	120	121	74	75	75	76	77	78	78
3 years														
Height (in)	35.8	36.4	37.3	38.4	39.6	40.6	41.2	35.8	36.4	37.3	38.4	39.6	40.6	41.2
Height (cm)	91.0	92.4	94.9	97.6	100.5	103.1	104.6	91.0	92.4	94.9	97.6	100.5	103.1	104.6
50th	88	89	89	90	91	92	93	48	48	49	50	51	53	53
90th	102	103	104	104	105	106	107	60	61	61	62	63	64	65
95th	106	106	107	108	109	110	110	64	65	65	66	67	68	69
95th + 12 mmHg	118	118	119	120	121	122	122	76	77	77	78	79	80	81
4 years														
Height (in)	38.3	38.9	39.9	41.1	42.4	43.5	44.2	38.3	38.9	39.9	41.1	42.4	43.5	44.2
Height (cm)	97.2	98.8	101.4	104.5	107.6	110.5	112.2	97.2	98.8	101.4	104.5	107.6	110.5	112.2
50th	89	90	91	92	93	94	94	50	51	51	53	54	55	55
90th	103	104	105	106	107	108	108	62	63	64	65	66	67	67
95th	107	108	109	109	110	111	112	66	67	68	69	70	70	71
95th + 12 mmHg	119	120	121	121	122	123	124	78	79	80	81	82	82	83
5 years														
Height (in)	40.8	41.5	42.6	43.9	45.2	46.5	47.3	40.8	41.5	42.6	43.9	45.2	46.5	47.3
Height (cm)	103.6	105.3	108.2	111.5	114.9	118.1	120.0	103.6	105.3	108.2	111.5	114.9	118.1	120.0
50th	90	91	92	93	94	95	96	52	52	53	55	56	57	57
90th	104	105	106	107	108	109	110	64	65	66	67	68	69	70
95th	108	109	109	110	111	112	113	68	69	70	71	72	73	73
95th + 12 mmHg	120	121	121	122	123	124	125	80	81	82	83	84	85	85
6 years														
Height (in)	43.3	44.0	45.2	46.6	48.1	49.4	50.3	43.3	44.0	45.2	46.6	48.1	49.4	50.3
Height (cm)	110.0	111.8	114.9	118.4	122.1	125.6	127.7	110.0	111.8	114.9	118.4	122.1	125.6	127.7
50th	92	92	93	94	96	97	97	54	54	55	56	57	58	59
90th	105	106	107	108	109	110	111	67	67	68	69	70	71	71
95th	109	109	110	111	112	113	114	70	71	72	72	73	74	74

95th + 12 mmHg	121	121	122	123	124	125	126	82	83	84	84	85	86	86
7 years														
Height (in)	45.6	46.4	47.7	49.2	50.7	52.1	53.0	45.6	46.4	47.7	49.2	50.7	52.1	53.0
Height (cm)	115.9	117.8	121.1	124.9	128.8	132.5	134.7	115.9	117.8	121.1	124.9	128.8	132.5	134.7
50th	92	93	94	95	97	98	99	55	55	56	57	58	59	60
90th	106	106	107	109	110	111	112	68	68	69	70	71	72	72
95th	109	110	111	112	113	114	115	72	72	73	73	74	74	75
95th + 12 mmHg	121	122	123	124	125	126	127	84	84	85	85	86	86	87
8 years														
Height (in)	47.6	48.4	49.8	51.4	53.0	54.5	55.5	47.6	48.4	49.8	51.4	53.0	54.5	55.5
Height (cm)	121.0	123.0	126.5	130.6	134.7	138.5	140.9	121.0	123.0	126.5	130.6	134.7	138.5	140.9
50th	93	94	95	97	98	99	100	56	56	57	59	60	61	61
90th	107	107	108	110	111	112	113	69	70	71	72	72	73	73
95th	110	111	112	113	115	116	117	72	73	74	74	75	75	75
95th + 12 mmHg	122	123	124	125	127	128	129	84	85	86	86	87	87	87
9 years														
Height (in)	49.3	50.2	51.7	53.4	55.1	56.7	57.7	49.3	50.2	51.7	53.4	55.1	56.7	57.7
Height (cm)	125.3	127.6	131.3	135.6	140.1	144.1	146.6	125.3	127.6	131.3	135.6	140.1	144.1	146.6
50th	95	95	97	98	99	100	101	57	58	59	60	60	61	61
90th	108	108	109	111	112	113	114	71	71	72	73	73	73	73
95th	112	112	113	114	116	117	118	74	74	75	75	75	75	75
95th + 12 mmHg	124	124	125	126	128	129	130	86	86	87	87	87	87	87
10 years														
Height (in)	51.1	52.0	53.7	55.5	57.4	59.1	60.2	51.1	52.0	53.7	55.5	57.4	59.1	60.2
Height (cm)	129.7	132.2	136.3	141.0	145.8	150.2	152.8	129.7	132.2	136.3	141.0	145.8	150.2	152.8
50th	96	97	98	99	101	102	103	58	59	59	60	61	61	62
90th	109	110	111	112	113	115	116	72	73	73	73	73	73	73
95th	113	114	114	116	117	119	120	75	75	76	76	76	76	76
95th + 12 mmHg	125	126	126	128	129	131	132	87	87	88	88	88	88	88
11 years														
Height (in)	53.4	54.5	56.2	58.2	60.2	61.9	63.0	53.4	54.5	56.2	58.2	60.2	61.9	63.0
Height (cm)	135.6	138.3	142.8	147.8	152.8	157.3	160.0	135.6	138.3	142.8	147.8	152.8	157.3	160.0
50th	98	99	101	102	104	105	106	60	60	60	61	62	63	64
90th	111	112	113	114	116	118	120	74	74	74	74	74	75	75
95th	115	116	117	118	120	123	124	76	77	77	77	77	77	77
95th + 12 mmHg	127	128	129	130	132	135	136	88	89	89	89	89	89	89
12 years														
Height (in)	56.2	57.3	59.0	60.9	62.8	64.5	65.5	56.2	57.3	59.0	60.9	62.8	64.5	65.5
Height (cm)	142.8	145.5	149.9	154.8	159.6	163.8	166.4	142.8	145.5	149.9	154.8	159.6	163.8	166.4
50th	102	102	104	105	107	108	108	61	61	61	62	64	65	65
90th	114	115	116	118	120	122	122	75	75	75	75	76	76	76
95th	118	119	120	122	124	125	126	78	78	78	78	79	79	79
95th + 12 mmHg	130	131	132	134	136	137	138	90	90	90	90	91	91	91
13 years														
Height (in)	58.3	59.3	60.9	62.7	64.5	66.1	67.0	58.3	59.3	60.9	62.7	64.5	66.1	67.0
Height (cm)	148.1	150.6	154.7	159.2	163.7	167.8	170.2	148.1	150.6	154.7	159.2	163.7	167.8	170.2

50th	104	105	106	107	108	108	109	62	62	63	64	65	65	66
90th	116	117	119	121	122	123	123	75	75	75	76	76	76	76
95th	121	122	123	124	126	126	127	79	79	79	79	80	80	81
95th + 12 mmHg	133	134	135	136	138	138	139	91	91	91	91	92	92	93
14 years														
Height (in)	59.3	60.2	61.8	63.5	65.2	66.8	67.7	59.3	60.2	61.8	63.5	65.2	66.8	67.7
Height (cm)	150.6	153.0	156.9	161.3	165.7	169.7	172.1	150.6	153.0	156.9	161.3	165.7	169.7	172.1
50th	105	106	107	108	109	109	109	63	63	64	65	66	66	66
90th	118	118	120	122	123	123	123	76	76	76	76	77	77	77
95th	123	123	124	125	126	127	127	80	80	80	80	81	81	82
95th + 12 mmHg	135	135	136	137	138	139	139	92	92	92	92	93	93	94
15 years														
Height (in)	59.7	60.6	62.2	63.9	65.6	67.2	68.1	59.7	60.6	62.2	63.9	65.6	67.2	68.1
Height (cm)	151.7	154.0	157.9	162.3	166.7	170.6	173.0	151.7	154.0	157.9	162.3	166.7	170.6	173.0
50th	105	106	107	108	109	109	109	64	64	64	65	66	67	67
90th	118	119	121	122	123	123	124	76	76	76	77	77	78	78
95th	124	124	125	126	127	127	128	80	80	80	81	82	82	82
95th + 12 mmHg	136	136	137	138	139	139	140	92	92	92	93	94	94	94
16 years														
Height (in)	59.9	60.8	62.4	64.1	65.8	67.3	68.3	59.9	60.8	62.4	64.1	65.8	67.3	68.3
Height (cm)	152.1	154.5	158.4	162.8	167.1	171.1	173.4	152.1	154.5	158.4	162.8	167.1	171.1	173.4
50th	106	107	108	109	109	110	110	64	64	65	66	66	67	67
90th	119	120	122	123	124	124	124	76	76	76	77	78	78	78
95th	124	125	125	127	127	128	128	80	80	80	81	82	82	82
95th + 12 mmHg	136	137	137	139	139	140	140	92	92	92	93	94	94	94
17 years														
Height (in)	60.0	60.9	62.5	64.2	65.9	67.4	68.4	60.0	60.9	62.5	64.2	65.9	67.4	68.4
Height (cm)	154.4	154.7	158.7	163.0	167.4	171.3	173.7	154.4	154.7	158.7	163.0	167.4	171.3	173.7
50th	107	108	109	110	110	110	111	64	64	65	66	66	66	67
90th	120	121	123	124	124	125	125	76	76	77	77	78	78	78
95th	125	125	126	127	128	128	128	80	80	80	81	82	82	82
95th + 12 mmHg	137	137	138	139	140	140	140	92	92	92	93	94	94	94

The 50th, 90th, and 95th percentiles were derived by using quantile regression on the basis of normal-weight children (BMI <85th percentile). BP stages are defined as elevated BP ≥90th percentile but <95th percentile; stage 1 HTN: ≥95th percentile or 130/80 to 139/89 mmHg; and stage 2 HTN: ≥95th percentile + 12 mmHg or >140/90 mmHg.

BMI: body mass index; BP: blood pressure; HTN: hypertension.

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Graphic 52646 Version 20.0

Table 6: **Pediatric respiratory rate and heart rate lower limit, normal range, and upper limit by age***

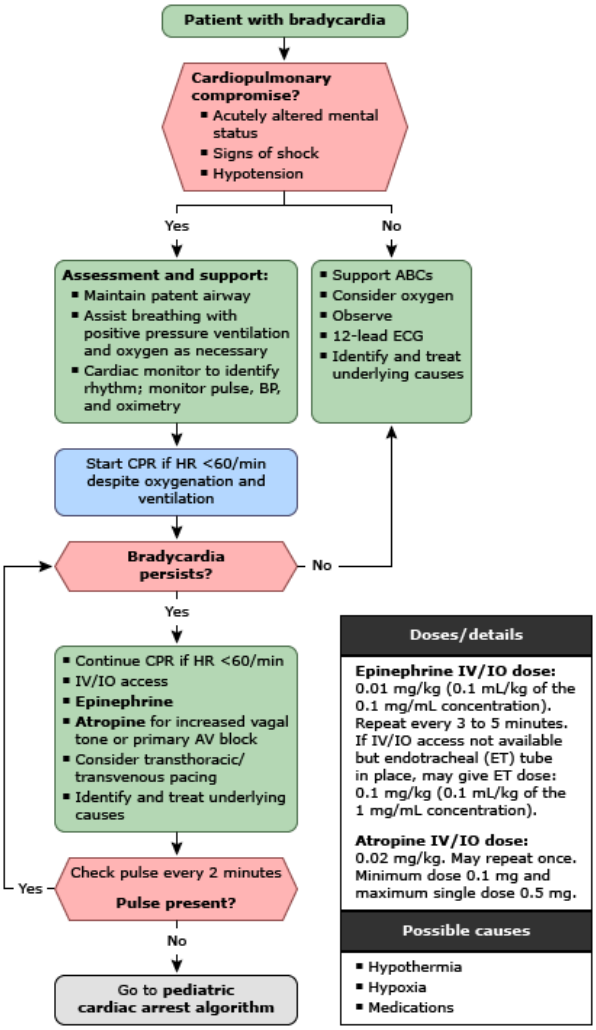
Age	Respiratory rate (breaths/minute)			Heart rate (beats/minute)		
	Lower limit (1st percentile)	Normal range (10th to 90th percentile)	Upper limit (99th percentile)	Lower limit (1st percentile)	Normal range (10th to 90th percentile)	Upper limit (99th percentile)
0 to 3 months	25	34 to 57	66	107	123 to 164	181
3 to <6 months	24	33 to 55	64	104	120 to 159	175
6 to <9 months	23	31 to 52	61	98	114 to 152	168
9 to <12 months	22	30 to 50	58	93	109 to 145	161
12 to <18 months	21	28 to 46	53	88	103 to 140	156
18 to <24 months	19	25 to 40	46	82	98 to 135	149
2 to <3 years	18	22 to 34	38	76	92 to 128	142
3 to <4 years	17	21 to 29	33	70	86 to 123	136
4 to <6 years	17	20 to 27	29	65	81 to 117	131
6 to <8 years	16	18 to 24	27	59	74 to 111	123
8 to <12 years	14	16 to 22	25	52	67 to 103	115
12 to <15 years	12	15 to 21	23	47	62 to 96	108
15 to 18 years	11	13 to 19	22	43	58 to 92	104

* The respiratory and heart rates provided are based upon measurements in awake, healthy infants and children at rest. Many clinical findings besides the actual vital sign measurement must be taken into account when determining whether a specific vital sign is normal in an individual patient. Values for heart rate or respiratory rate that fall within normal limits for age may still represent abnormal findings that are caused by underlying disease in a particular infant or child.

Data from: Fleming S, Thompson M, Stevens R, et al. Normal ranges of heart rate and respiratory rate in children from birth to 18 years of age: A systematic review of observational studies. Lancet 2011; 377:1011.

Graphic 78097 Version 11.0

Algorithm 3: **Pediatric bradycardia with a pulse 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Bradycardia with a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ABCs: airway, breathing, circulation; AV: atrioventricular; BP: blood pressure; CPR: cardiopulmonary resuscitation; ECG: electrocardiogram; ET: endotracheal; HR: heart rate; IO: intraosseous; IV: intravenous.

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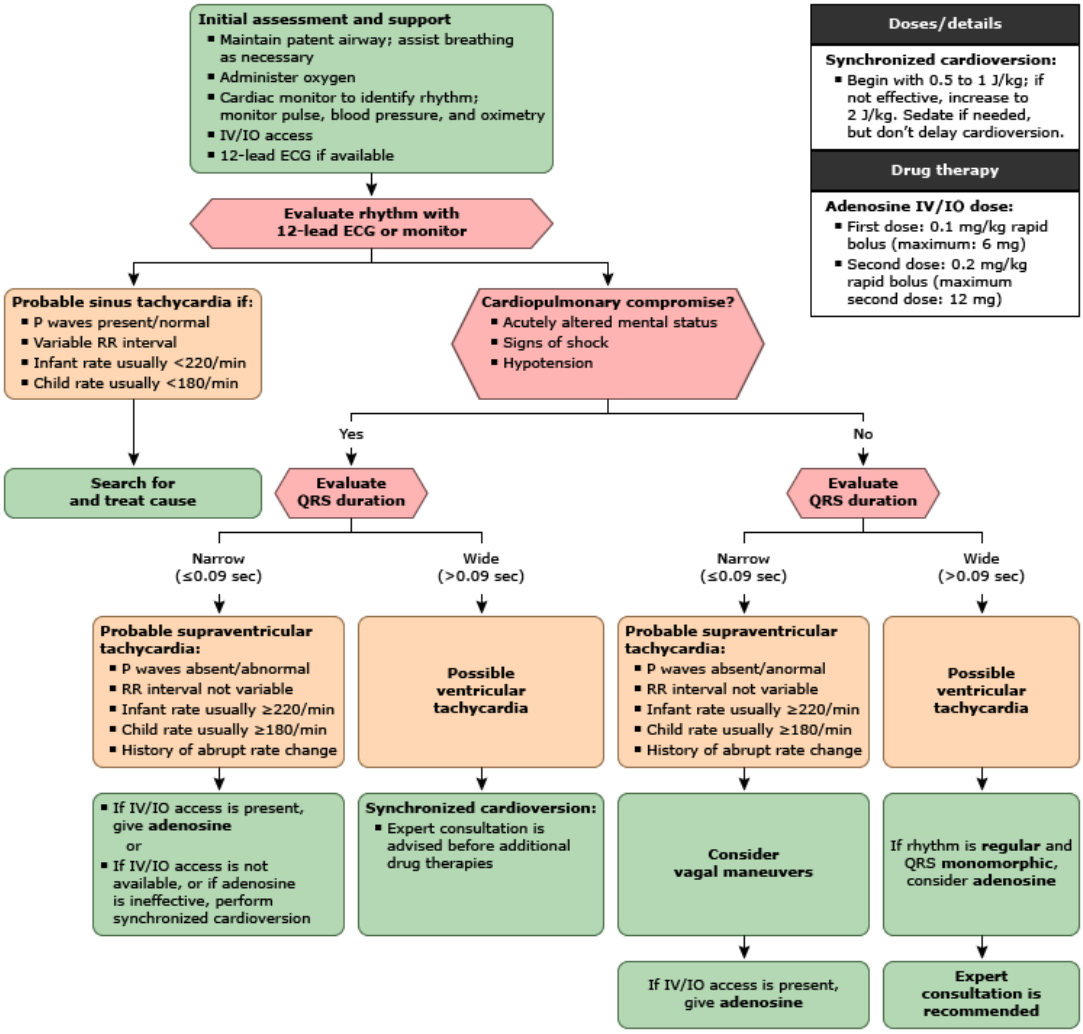
Graphic 129941 Version 11.0

Algorithm 4: **Pediatric tachycardia with a pulse 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Tachyarrhythmia with a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ECG: electrocardiogram; IO: intraosseous; IV: intravenous; J/kg: joules per kilogram.

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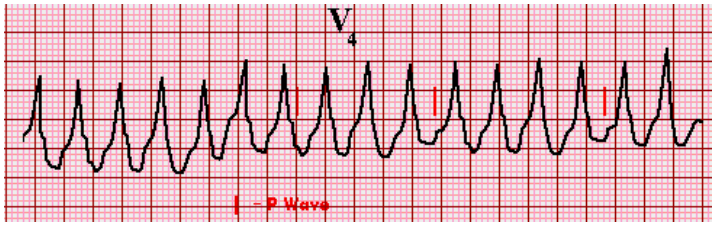
Graphic 129940 Version 12.0

Waveform 1: **Atrioventricular reentrant tachycardia**

Atrioventricular reentrant tachycardia breaking to sinus rhythm with Wolff-Parkinson-White syndrome.

Graphic 77867 Version 3.0

Waveform 2: **Single lead electrocardiogram (ECG) showing monomorphic ventricular tachycardia**



Three or more successive ventricular beats are defined as ventricular tachycardia (VT). This VT is monomorphic since all of the QRS complexes have an identical appearance. Although the P waves are not distinct, they can be seen altering the QRS complex and ST-T waves in an irregular fashion, indicating the absence of a relationship between the P waves and the QRS complexes (ie, AV dissociation is present).

AV: atrioventricular.

Graphic 63176 Version 7.0

Normal rhythm strip

Normal rhythm strip in lead II. The PR interval is 0.15 sec and the QRS duration is 0.08 sec. Both the P and T waves are upright.

Courtesy of Morton F Arnsdorf, MD.

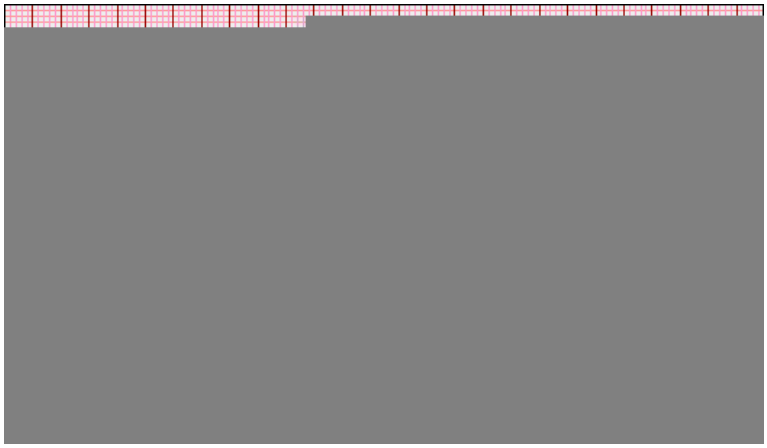
Graphic 59022 Version 3.0

Table 7: **Drug- and toxin-induced electrocardiographic abnormalities**

Bradycardia/AV blockade	Supraventricular tachycardia	Ventricular tachycardia	QRS and QT interval prolongation
Beta blockers	Sympathomimetics	Sympathomimetics	Antidepressants
Calcium channel blockers	Amphetamines	Cocaine	Citalopram
Cardiac glycosides	Cocaine	Amphetamines	Escitalopram
Digoxin	Theophylline	Theophylline	Bupropion
Digitoxin	Caffeine	Antidepressants	Venlafaxine
Red squill	Methylphenidate	TCAs	Desvenlafaxine
Digitalis lanata	Ephedrine	Antipsychotics	Tricyclic antidepressants
Digitalis purpurea	Pseudoephedrine	Phenothiazines	Antipsychotics
Bufotenin	Albuterol	Chlorinated hydrocarbons	Antihistamines
Oleander	Dobutamine	Chloral hydrate	Diphenhydramine
Alpha-adrenergic agonists	Epinephrine	Solvents	Astemizole
Phenylpropanolamine	Dopamine	Fluoride	Terfenadine
Clonidine	Anticholinergics	Cardiac glycosides	Antiarrhythmics
Imidazolines	Antihistamines	Potassium	Quinidine
Cholinergics	TCAs		Disopyramide
Organophosphates	Phenothiazines		Procainamide
Carbamates	Clozapine		Propafenone
Opioids	Atropine		Flecainide, encainide
Sedative-hypnotics	Scopolamine		Amiodarone
Magnesium	Thyroid hormone		Calcium channel blockers (rare)
	Cellular asphyxiants		Beta blockers (rare)
	Carbon monoxide		Ondansetron
	Drug withdrawal states		Propoxyphene
			Organophosphate insecticides
			Antimicrobials
			Amantadine
			Azithromycin
			Chloroquine
			Erythromycin
			Pentamidine
			Quinine
			Quinolones (eg, ciprofloxacin)
			Arsenic
			Thallium
			Fluoride
			Citrate
			Lithium

Graphic 66773 Version 10.0

Waveform 3: **Continuous electrocardiographic (ECG) strip during an episode of ventricular fibrillation (VF) that progresses to fine VF and then asystole**



At the onset of ventricular fibrillation (VF), the QRS complexes are regular, widened, and of tall amplitude, suggesting a more organized ventricular tachyarrhythmia. Over a brief period of time, the rhythm becomes more disorganized with high amplitude fibrillatory waves; this is coarse VF. After a longer period of time, the fibrillatory waves become fine and appear similar to asystole.

Graphic 67777 Version 4.0

Waveform 4: **Nonsustained torsades de pointes**

Nonsustained runs of polymorphic ventricular tachycardia in a patient with prolonged QTc in the setting of pneumonia treated with azithromycin, stress (takotsubo) cardiomyopathy, and hypokalemia.

QTc: corrected QT interval.

Graphic 144095 Version 1.0

Algorithm 5: **Pediatric cardiac arrest algorithm for suspected or confirmed COVID-19 patients**

Pediatric cardiac arrest algorithm for patients with suspected or confirmed COVID-19.

PPE: personal protective equipment; CPR: cardiopulmonary resuscitation; HEPA: high efficiency particulate air; VF: ventricular fibrillation; pVT: pulseless ventricular tachycardia; PEA: pulseless electrical activity; IV: intravenous; IO: intraosseous; ROSC: return of spontaneous circulation; AGP: aerosol-generating procedure; ET: endotracheal.

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Graphic 127842 Version 12.0

Picture 1: **Two rescuer bag-mask ventilation**

Two rescuers may be necessary to provide ventilation. One rescuer may need both hands to open the airway and make the seal between the child's face and the mask. The second rescuer is then needed to provide positive pressure ventilation.

Graphic 72516 Version 5.0

Table 8: **Initial settings for volume-controlled mechanical ventilation in children***

	Infant (<1 year of age)	**Toddler/child (1 to 12 years)**	**Adolescent (>12 years)**
Tidal volume (mL)	5 to 8 mL/kg (healthy lungs)	5 to 8 mL/kg (healthy lungs)	5 to 8 mL/kg (healthy lungs)
	3 to 6 mL/kg (lung protective strategy)	3 to 6 mL/kg (lung protective strategy)	3 to 6 mL/kg (lung protective strategy)
Rate (breaths/minute)	20 to 30	15 to 25	12 to 20
PEEP (cm H_2O)	4 to 8	4 to 8	4 to 8
Pressure support (cm H_2O)¶	Minimum 6 to 10	Minimum 6 to 10	Minimum 6 to 10
Peak inspiratory flow (L/minute)	Adjusted to fit desired inspiratory time	Adjusted to fit desired inspiratory time	Adjusted to fit desired inspiratory time
Inspiratory time (seconds) Targeted, based upon changes to inspiratory flow	0.4 to 0.6	0.7 to 0.9	0.9 to 1.2
FiO_2 (%)Δ	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6
Flow trigger (L/minute)	0.25 to 0.5	0.8 to 2	0.8 to 2
Pressure support cycle	10 to 25% of peak flow rate	10 to 25% of peak flow rate	10 to 25% of peak flow rate

FiO_2: fraction of inspired oxygen; PEEP: positive end-expiratory pressure.

* Consultation with an expert in the mechanical ventilation of children (eg, pediatric intensivist or pediatric anesthesiologist) is strongly encouraged. Regardless of ventilator settings employed, the physician must assess ventilator settings shortly after initiation and frequently thereafter and adjust them as needed to meet oxygenation and ventilation goals as the natural course of the underlying pathophysiology evolves.

¶ Adjust according to endotracheal tube size: 3 to 3.5 mm: 10 cm H_2O; 4 to 4.5 mm: 8 cm H_2O; ≥5 mm: 6 cm H_2O.

Δ Wean to 0.6 or below to maintain arterial oxygen tension (PaO_2) 60 to 80, oxygen saturation (SpO_2) 92 to 97% when required PEEP is <10 cm H_2O; if required PEEP is ≥10 cm H_2O, then targeted oxygen saturation may be reduced to 88 to 92% for those patients with pediatric acute respiratory distress syndrome (PARDS).

Graphic 113548 Version 4.0

Table 9: **Initial settings for pressure-controlled mechanical ventilation in children** *

	Infant (<1 year of age)	**Toddler/child (1 to 12 years)**	**Adolescent (>12 years)**
PIP (cm H_2O)¶	16 to 25	16 to 25	16 to 25
PEEP (cm H_2O)¶	4 to 8	4 to 8	4 to 8
Rate (breaths/minute)	20 to 30	15 to 25	12 to 20
Inspiratory time (seconds)	0.4 to 0.6	0.7 to 0.9	0.9 to 1.2
Pressure support (cm H_2O)Δ	Minimum 6 to 10	Minimum 6 to 10	Minimum 6 to 10
FiO_2 (%)◇	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6
Flow trigger (L/minute)	0.25 to 0.5	0.8 to 2	0.8 to 2
Pressure support cycle	10 to 25% of peak flow rate	10 to 25% of peak flow rate	10 to 25% of peak flow rate

FiO_2: fraction of inspired oxygen; PEEP: positive end-expiratory pressure; PIP: peak inspiratory pressure.

* Consultation with an expert in the mechanical ventilation of children (eg, pediatric intensivist or pediatric anesthesiologist) is strongly encouraged. Regardless of ventilator settings employed, the physician must assess ventilator settings shortly after initiation and frequently thereafter and adjust them as needed to meet oxygenation and ventilation goals as the natural course of the underlying pathophysiology evolves.

¶ Initial pressure settings can be established based upon pressures identified during manual ventilation with a bag and a manometer/standalone respiratory mechanics monitor prior to switching to the ventilator.

Δ Adjust according to endotracheal tube size: 3 to 3.5 mm: 10 cm H_2O; 4 to 4.5 mm: 8 cm H_2O; ≥5 mm: 6 cm H_2O.

◇ Wean to 0.60 or below to maintain arterial oxygen tension (PaO_2) 60 to 80, oxygen saturation (SpO_2) 92 to 97% when required PEEP is <10 cm H_2O; if required PEEP is ≥10 cm H_2O then targeted oxygen saturation may be reduced to 88 to 92% for those patients with pediatric acute respiratory distress syndrome (PARDS).

Graphic 113549 Version 3.0

Table 10: **Pharmacologic agents for pediatric advanced life support**

Agent	Indications	Dose
Adenosine*	▪ Supraventricular tachycardia (SVT)	▪ Initial dose 0.1 mg/kg (children >50 kg receive 6 mg) given as rapid IV or IO push closest to central circulation; follow immediately with 3 to 5 mL saline flush (10 to 20 mL saline flush for larger child or adolescent). ▪ If not responsive in 2 minutes, give second dose of 0.2 mg/kg (children >50 kg receive 12 mg), follow immediately with 5 mL saline flush; if not responsive after additional 2 minutes, give third dose of 0.3 mg/kg (maximum 12 mg) follow immediately with 5 mL saline flush.
Amiodarone¶	▪ Pulseless ventricular arrhythmias not responsive to CPR, defibrillation, and epinephrine ▪ Stable ventricular tachycardia ▪ SVT refractory to adenosine	▪ **Cardiac arrest:** 5 mg/kg rapid (eg, over ~3 minutes) IV or IO bolus (maximum dose 300 mg); may repeat 5 mg/kg dose (maximum dose 150 mg) twice up to a total of 15 mg/kg or 600 mg, whichever is less) during acute treatment. ▪ **Perfusing patient:** 5 mg/kg IV or IO (maximum dose 300 mg) dilute to 2 mg/mL or less and infuse over 20 to 60 minutes; may repeat 5 mg/kg dose (maximum dose 150 mg) twice up to a total of 15 mg/kg or 600 mg, whichever is less up to 3 doses during acute treatment.
Atropine	▪ Vagally mediated bradycardia ▪ Primary atrioventricular block ▪ Bradycardia not responsive to oxygen, airway support, and epinephrine administration ▪ Prevention of bradycardia during endotracheal intubation for children <1 year of age, children with septic or late-stage hypovolemic shock, children 1 to 5 years of age receiving succinylcholine, and children over 5 years of age receiving a second dose of succinylcholine	▪ 0.02 mg/kg IV or IO[Δ]; maximum single dose 0.5 mg for child or 1 mg for adolescent); may repeat once in 3 to 5 minutes. ▪ Maximum total dose 1 mg (child) or 2 mg (adolescent).[◇] ▪ Only if IV and IO not available, may give via endotracheal tube (ET) 0.04 to 0.06 mg/kg diluted with 3 to 5 mL saline or undiluted followed by flush; repeat once if needed (IV or IO are preferred).
Calcium chloride§	▪ Not routinely recommended without 1 of the following conditions: • Hypocalcemia • Hypermagnesemia • Hyperkalemia • Calcium channel blocker (CCB) overdose	▪ **Cardiac arrest:** 20 mg/kg given as 0.2 mL/kg of 10% calcium chloride solution, maximum 2 g (20 mL, 14 mmol) per dose; dilute in an equal amount of D5W or NS and give slow IV or IO push; repeat in 10 minutes if needed. ▪ **Perfusing patient:** 10 to 20 mg/kg given as 0.1 to 0.2 mL/kg of 10% calcium chloride solution, maximum 2 g (20 mL, 14 mmol) per dose; dilute in an equal amount of D5W or NS and give IV or IO over 5 to 10 minutes (for initial stabilization in CCB overdose or severe symptomatic hyperkalemia); or dilute in 10 to 50 mL D5W or NS and give IV or IO over 10 to 20 minutes in perfusing patient with severe symptomatic hypocalcemia (eg, tetany, carpopedal spasm, or seizure); may repeat in 10 minutes if needed. ▪ Conversions: 10% calcium chloride = 100 mg/mL calcium chloride = 27.3 mg/mL elemental calcium = 0.68 mmol/mL calcium. ▪ Peripheral administration of calcium chloride is not recommended.[§] ▪ Calcium formulations precipitate with sodium bicarbonate; do not co-infuse.
Calcium gluconate	▪ Not routinely recommended without 1 of the following conditions: • Hypocalcemia • Hypermagnesemia • Hyperkalemia • Calcium channel blocker (CCB) overdose • (Alternate to calcium chloride for perfusing patient; provides slower correction of ionized calcium concentration but less likely to	▪ **Cardiac arrest:** 60 to 100 mg/kg given as 0.6 to 1 mL/kg of 10% calcium gluconate solution, maximum 3 g (30 mL, 7 mmol) per dose; dilute in an equal amount of D5W or NS and give slow IV or IO push; repeat in 10 minutes if needed. Calcium chloride preferred if available and patient has central venous access. ▪ **Perfusing patient:** 60 to 100 mg/kg given as 0.6 mL/kg of 10% calcium gluconate solution, maximum 2 g (20 mL, 4.5 mmol) per dose; dilute in an equal amount of D5W or NS and give IV or IO over 5 to 10 minutes for initial stabilization in CCB overdose or severe symptomatic hyperkalemia, may repeat if needed or follow by continuous infusion (CCB overdose); or dilute in 10 to 50 mL D5W or

	cause tissue injury when given by peripheral IV)	NS and give IV or IO over 10 to 20 minutes in perfusing patient with severe symptomatic hypocalcemia (eg, tetany, carpopedal spasm, or seizures); may repeat in 10 minutes if needed. ▪ Conversions: 10% calcium gluconate = 100 mg/mL calcium gluconate = 9.3 mg/mL elemental calcium = 0.23 mmol/mL calcium. ▪ Calcium formulations precipitate with sodium bicarbonate; do not co-infuse.
Epinephrine[¥]	▪ Asystole ▪ Pulseless electrical activity ▪ Pulseless ventricular arrhythmias not responsive to initial defibrillation ▪ Bradycardia with poor perfusion not responsive to oxygenation, ventilation, and requiring cardiac compressions (heart <60 beats/minute)	▪ 0.01 mg/kg IV or IO given as 0.1 mL/kg using the 0.1 mg/mL solution up to 1 mg per dose. ▪ Repeat every 3 to 5 minutes as needed (eg, until return of spontaneous circulation); only if IV and IO are not available, may give endotracheal (ET) 0.1 mg/kg as 0.1 mL/kg using the 1 mg/mL solution up to 2.5 mg per dose diluted to 3 to 5 mL with saline or undiluted followed by 3 to 5 mL flush; repeat every 3 to 5 minutes as needed (eg, until return of spontaneous circulation or IV/IO access established). ▪ Epinephrine is inactivated by sodium bicarbonate. ▪ Examples of epinephrine infusion for refractory anaphylaxis (perfusing patient) are provided as separate tables in UpToDate.
Glucose (dextrose)	▪ Documented blood glucose ≤60 mg/dL (3.3 mmol/L)[‡]	▪ 0.5 to 1 g/kg, IV or IO, as follows: • Infants and children <5 years: 5 to 10 mL/kg of 10% dextrose solution. • Children ≥5 years: 2 to 4 mL/kg of 25% dextrose solution (preferred when available) or 1 to 2 mL/kg of 50% dextrose solution.
Lidocaine	▪ Pulseless ventricular arrhythmias not responsive to CPR, defibrillation, and epinephrine	▪ 1 mg/kg rapid IV or IO bolus. ▪ Follow the bolus with an infusion of 20 to 50 mcg/kg/minute. If the start of the infusion will be delayed longer than 15 minutes, then a second IV or IO bolus dose of 1 mg/kg is suggested. ▪ Only if IV and IO not available, may give via endotracheal tube (ET) 2 to 3 mg/kg, flush with 5 mL NS, and follow with 5 assisted manual ventilations (IV and IO are preferred).
Magnesium sulfate	▪ Polymorphic ventricular tachycardia (torsades de pointes) ▪ Documented hypomagnesemia	▪ **Cardiac arrest** (pulseless torsades)**:** 25 to 50 mg/kg; given as 0.05 to 0.1 mL/kg of 50% magnesium sulfate solution up to maximum 2 g (4 mL) per dose; dilute in 10 mL D5W, give IV or IO over 1 to 2 minutes. ▪ **Perfusing patient** (torsades, hypomagnesemia, status asthmaticus)**:** Same dose as for cardiac arrest, except dilute dose in 10 to 50 mL D5W or NS and infuse over 15 to 20 minutes (maximum 150 mg per minute).[†] ▪ Conversions: 50% magnesium sulfate = 500 mg/mL magnesium sulfate = 2 mmol/mL magnesium.
Oxygen	▪ All infants and children outside of the neonatal period	▪ 100% initial dose via nonrebreathing face mask or bag-mask ventilation, wean as clinically indicated to maintain pulse oximetry 94 to 99%.
Procainamide[**]	▪ Stable ventricular tachycardia ▪ SVT in patients with Wolff-Parkinson-White syndrome or unstable SVT refractory to adenosine and cardioversion	▪ Loading dose (pediatric cardiology consultation advised): • Neonates: 7 to 10 mg/kg IV or IO. • Older infants and children ≥1 year: 15 mg/kg IV or IO. • Maximum: 1 g. • To avoid transient hypotension caused by rapid administration, give the loading dose slowly over 30 to 60 minutes. During loading, ensure frequent BP measurements and continuous ECG monitoring. • For stable patients in normal sinus rhythm who are receiving procainamide, stop administration if the QRS interval increases >50% from baseline or an arrhythmia develops. ▪ After the loading dose, start a continuous IV infusion at 20 mcg/kg per minute and titrate up to a maximum dose of 80 mcg/kg per minute, as needed, for rhythm control (maximum daily dose, 2 g over 24 hours). ▪ Measure plasma levels (procainamide and N-acetyl procainamide) 4 hours after completion of the loading dose.

Sodium bicarbonate¶¶	▪ Not routinely recommended without 1 of the following conditions: • Hyperkalemia • Poisoning by sodium channel blocking agents (eg, cyclic antidepressants, type Ia antiarrhythmic agents) with prolongation of QRS interval (>0.1 msec) • Prolonged cardiac arrest with documented severe metabolic acidosis (routine use in resuscitation is **not** recommended) • Shock with documented metabolic acidosis	▪ Infants <6 months: 1 mEq/kg IV or IO given as 2 mL/kg of 4.2% solution. ▪ Infants ≥6 months and children: 1 mEq/kg IV or IO given as 1 mL/kg of 8.4% solution. ▪ Maximum single dose 50 mEq (child) to 100 mEq (adolescent). ▪ 0.5 mEq/kg subsequent doses after 10 minutes given as: • Child: 0.5 mL/kg of 8.4% solution. • Infants under 6 months: 1 mL/kg of 4.2% solution. ▪ Forms precipitate with calcium and can inactivate epinephrine; do not co-infuse.
Sotalol∆∆	▪ Unstable SVT refractory to adenosine and cardioversion	▪ Loading dose (pediatric cardiology consultation advised): • Initial loading dose: 1 mg/kg (maximum 80 mg/dose) IV over 30 to 60 minutes. • BSA-directed dosing and age-based dosing also reported. • Prior to starting maintenance dosing, obtain an ECG and check for QTc ≥0.50 msec in patients with a normal baseline QTc or, in patients with bundle branch block at baseline, significant prolongation. ▪ Correct abnormalities of serum potassium and magnesium prior to administration (target serum potassium >4 mEq/L, serum magnesium >1.5 mEq/L). ▪ May cause severe and potentially fatal ventricular arrhythmias, primarily torsade de pointes, but also sustained ventricular tachycardia or ventricular fibrillation; bradycardia and hypotension; or worsening heart failure in patients with depressed left ventricular function. Multiple contraindications to co-administration with other medications related to nonselective beta-blocking and cardiac action potential prolonging effects. ▪ Subsequent dose and dosing interval must be adjusted in infants and children ≤2 years according to an age-related factor obtained from the manufacturer nomogram; refer to drug monograph.

Recommendations in this table are generally consistent with prior Pediatric Advanced Life Support (PALS) guidelines[1,2]. Details concerning dilution and administration of emergency drugs are based upon recommendations used at experienced pediatric centers. Protocols vary. The use of a stopcock is preferred for IV emergency medications because it permits rapid, sequential drug delivery without the need to disconnect lines, thereby reducing the risk of contamination. For additional details, refer to the UpToDate topic on primary drugs in pediatric resuscitation and the LexiDrug monographs for specific drugs.

BP: blood pressure; CCB: calcium channel blocker; CPR: cardiopulmonary resuscitation; ECG: electrocardiogram; ET: endotracheally; IO: intraosseous; IV: intravenous; SVT: supraventricular tachycardia.

* May cause atrial fibrillation with progression to ventricular fibrillation in children with Wolff-Parkinson-White syndrome. As a result, caution should be used when giving adenosine if Wolff-Parkinson-White is a possible mechanism; emergency resuscitation equipment should be available, and when possible, a pediatric cardiologist should be consulted before treatment. NOTE: Doses <1 mg may require dilution (eg, in 1 mL normal saline).

¶ Amiodarone should not be administered with other drugs that may cause QT prolongation (eg, procainamide or sotalol) without cardiology consultation or to patients with congenital long QT syndrome. Cardiology consult is advised prior to use when patient has a perfusing rhythm.

∆ When used to treat bradycardia with poor perfusion in patients weighing less than 5 kg, a minimum atropine dose of 0.1 mg may be considered[2]. However, many centers prefer using the weight-based dose without a minimum dose in this situation.

◇ Patients with poisoning from cholinesterase-inhibiting agents may require much higher doses of atropine to dry bronchial secretions. Refer to UpToDate topics on organophosphate and carbamate poisoning.

§ Rapid administration can cause bradycardia or asystole. Calcium chloride should be diluted and given through central venous or intraosseous access, if possible, to avoid potential tissue necrosis or sloughing in the event of extravasation.

¥ To help prevent medication errors, ratio expressions have been removed from epinephrine labels in the United States. Ampules, vials, and syringes of epinephrine with ratio expressions may, however, remain available in some other regions. The 0.1 mg/mL concentration of epinephrine may be labeled as 1:10,000 and the 1 mg/mL concentration may be labeled as 1:1000.

‡ Routine administration of glucose without evaluation of the serum glucose is not recommended. Empiric treatment with glucose may be appropriate if bedside glucose determination is not available and the infant or child has symptoms of hypoglycemia or is at risk for

developing hypoglycemia. Lower doses of glucose (eg, 0.25 g/kg or 2.5 mL/kg of 10% dextrose solution) have been proposed by some experts to avoid osmotic diuresis and insulin surge with recurrent hypoglycemia. Refer to UpToDate topics on hypoglycemia in infants and children.

† Rapid infusions of magnesium sulfate in perfusing patients are associated with hypotension and asystole.

** May be used safely in children with Wolff-Parkinson-White syndrome. Should not be used in patients who have received other drugs that prolong the QT interval (eg, amiodarone or sotalol) without cardiology consultation or in patients with congenital prolonged QT syndrome.

¶¶ Sodium bicarbonate should only be administered to children with adequate ventilation. Flush well with normal saline before and after administration to avoid alkaline inactivation of epinephrine or precipitation with calcium-containing solutions.

ΔΔ Should not be used in patients who have received other drugs that prolong the QT interval (eg, amiodarone or procainamid) without cardiology consultation.

Courtesy of Pamela Bailey, MD and Susan B Torrey, MD with additional data from:

1. *Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: Pediatric Advanced Life Support (PALS): 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2010; 122(18 Suppl 3):S876 as updated by de Caen A, Berg M, Chameides L et al. Part 12: Pediatric Advanced Life Support (PALS): 2015 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2015; 132(suppl 2):S526.*
2. *Lasa JJ, Dhillon GS, Duff JP, Hayes J, et al. Part 8: Pediatric Advanced Life Support: 2025 American Heart Association and American Academy of Pediatrics Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2025; 152(suppl 2):S479.*

Graphic 70539 Version 25.0

Algorithm 6: **Pediatric cardiac arrest algorithm 2020 update**

Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Cardiac Arrest Algorithm is now available. This algorithm will be updated within UpToDate in the future.

CPR: cardiopulmonary resuscitation; ET: endotracheal; IO: intraosseous; IV: intravenous; PEA: pulseless electrical activity; pVT: pulseless ventricular tachycardia; ROSC: return of spontaneous circulation; VF: ventricular fibrillation.

* If ROSC, go to post-cardiac arrest care checklist. The post-cardiac arrest care checklist can be found here.

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Graphic 129942 Version 12.0

Contributor Disclosures

Eric Fleegler, MD, MPH No relevant financial relationship(s) with ineligible companies to disclose. **Monica Kleinman, MD** Consultant/Advisory Boards: American Heart Association [Pediatric resuscitation]; International Liaison Committee on Resuscitation [Resuscitation]. All of the relevant financial relationships listed have been mitigated. **Susan B Torrey, MD** No relevant financial relationship(s) with ineligible companies to disclose. **James F Wiley, II, MD, MPH** No relevant financial relationship(s) with ineligible companies to disclose.

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